CareLoop-Edge

Assistive AI follow-up and continuity of care for brain MRI on a Raspberry Pi

Project report

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Repository	`https://github.com/TanmayRawal/CareloopEdge`
Release described	model `mri/0.3.0`, portal commit `2c90ba2`
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Research prototype on public data. It is not a medical device and is not intended for clinical decisions. Every AI result in the system is reviewed and signed by a radiologist before a patient sees anything.

Abstract

CareLoop-Edge is a brain-MRI decision-support and follow-up system that runs entirely on a Raspberry Pi in a clinic. A patient, or a health worker on their behalf, uploads a single image or a whole MRI study. A quantised ResNet-50 classifier with a segmentation head analyses it on the device. The result goes to a radiologist's worklist with calibrated probabilities, a 95% prediction set, an exact lesion-focused Grad-CAM explanation and the lesion outline. The radiologist signs the report and chooses the next step (consultation, repeat MRI or referral). The care-loop part of the system then makes sure that step happens: appointment booking, reminders, a risk-ranked follow-up worklist for the coordinator, referral tracking and analytics on loop closure.

The model is trained on a PC, audited for data leakage, quantised to INT8, and reaches the device as a signed and encrypted bundle from a self-hosted registry. Patient data is encrypted at rest and every access is written to a tamper-evident audit log. After feedback from a radiologist that diagnosis is made from the whole study, the model was evaluated on complete 3D patient scans, retrained, and released as `mri/0.3.0`. On held-out patients it reaches a tumour-type macro-F1 of 0.98, a slice-level tumour AUROC of 0.95 and a 5% false-alarm rate on healthy volunteers, meeting all three pre-registered targets.

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1 Problem and goals

In district hospitals and screening camps a brain MRI may be acquired, read once, and then the patient is lost. The nearest neurosurgeon or neuroradiologist can be 60–100 km away, and nobody tracks whether the follow-up scan, the consultation or the referral ever happened. The project addresses two linked problems:

1. **Reading support at the edge.** Give the radiologist a fast, explainable second look at the study, computed on a low-cost device in the clinic, with no cloud GPU and no patient data leaving the site.
2. **Closing the care loop.** After the report is signed, track the patient until the next step chosen by the radiologist has actually happened, and make the patients at risk of being lost visible to the care team.

Design principles that run through the whole system: the AI assists and the doctor decides; patients never see raw AI output; every number shown is measured, not estimated; and the system keeps working offline.

2 Project workflow

Figure 1 summarises the order in which the work was done. Each stage produced a measured result recorded in `docs/FINDINGS.md`; several stages changed the plan of the next one.

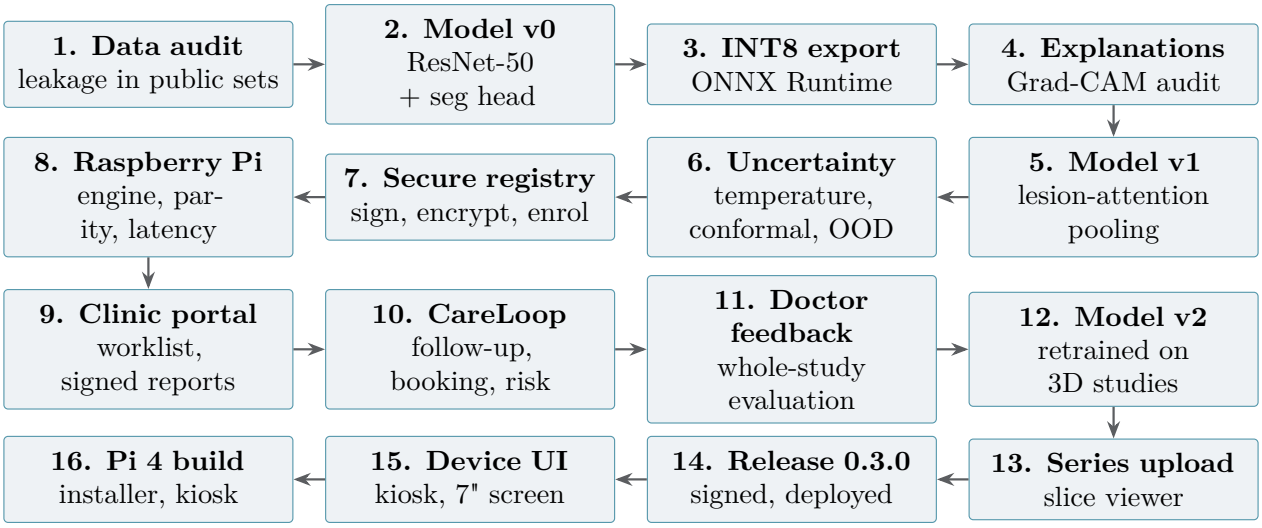


Figure 1: Order of the work. Stages 1–8 build and secure the model; 9–10 build the clinical product; 11–14 respond to a radiologist's feedback that diagnosis is made from the whole study; 15–16 turn it into a stand-alone device.

1. **Data audit.** Public brain-tumour MRI benchmarks were found to leak heavily: 566 of the 1,000 official BRISC test images are copies (462 mirrored) of training images, and other public sets largely duplicate BRISC. A flip-aware similarity search defines leakage-free, patient-grouped splits used for every result in this report.
2. **Model v0.** A ResNet-50 classifier (glioma, meningioma, pituitary tumour, no tumour) with an FPN segmentation head, trained on the leakage-free split.
3. **INT8 export.** ONNX export and static INT8 quantisation (96.6 MB to 24.7 MB). The audit showed that v0 INT8 silently lost 11 macro-F1 points on external data; external INT8 evaluation became part of acceptance.

4. **Explanations.** Grad-CAM, LayerCAM, Finer-CAM and ShapleyCAM were evaluated against expert tumour masks. The v0 maps put only 15% of their evidence inside the tumour, and pituitary maps highlighted anatomy, not tumour.
5. **Model v1.** A lesion-attention pooling head makes the class decision lesion-focused by construction while keeping Grad-CAM exact (Section 5). Evidence inside the tumour rose from 0.15 to 0.73.
6. **Uncertainty.** Temperature scaling, a class-conditional conformal 95% prediction set and a Mahalanobis out-of-distribution detector, all fitted on the deployed INT8 graph.
7. **Secure model registry.** Models are signed (OpenSSF Model Signing) and encrypted per device; a 16-point end-to-end security check passes against the live server.
8. **Raspberry Pi.** The engine runs with numpy, OpenCV and ONNX Runtime only; Pi and PC agree on 40/40 scans; a full analysis takes 238 ms.
9. **Clinic portal.** Patients upload, radiologists review and sign, patients read the signed report.
10. **CareLoop.** Follow-up plans, booking, reminders, a contact log, referrals and an explainable risk score.
11. **Doctor feedback.** A radiologist pointed out that a diagnosis is never made from one image. The deployed model was therefore tested on complete 3D studies against pre-registered targets, and failed all three.
12. **Model v2.** Retrained with slices from 3D patient studies (BraTS 2023 glioma and meningioma, NHNN pituitary); all three targets met on held-out patients.
13. **Whole-series upload.** The portal accepts NIfTI files, ZIP archives and multi-file DICOM series; the doctor scrolls through 36 slices in three planes with Grad-CAM on the three key slices.
14. **Release 0.3.0.** Published through the registry and verified on the Pi.
15. **Device interface.** A clinical workstation design system, a device home screen with a 3D brain reconstructed from a real MRI, a touch keyboard and a kiosk mode for the official 7-inch display.
16. **Pi 4 build.** A second device installed from a freshly written SD card with the same installer.

3 System architecture

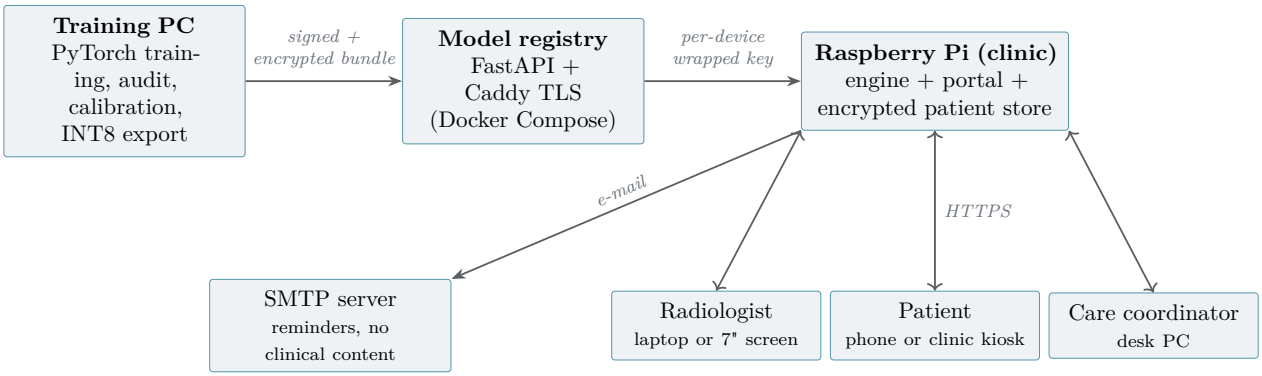


Figure 2: Components. The model leaves the training PC only as a signed, encrypted bundle. Patient data never leaves the clinic device; outgoing e-mails carry no clinical content.

3.1 Technology stack

Layer	Technology and role
Model training	PyTorch, timm (ResNet-50, ImageNet init), AMP bf16, EMA weights, AdamW, cosine schedule
Data handling	NumPy, OpenCV, pydicom, nibabel (NIfTI), SciPy; flip-aware NCC deduplication; pandas manifests
Evaluation	scikit-learn metrics, bootstrap confidence intervals, McNemar tests, CAM faithfulness (pointing, ROAD)
Export and inference	ONNX, ONNX Runtime with static INT8 quantisation (QDQ), CPU only, no PyTorch on the device
Uncertainty	temperature scaling, class-conditional conformal prediction, multi-layer Mahalanobis OOD
Model security	OpenSSF Model Signing (ECDSA P-256), AES-256-GCM, X25519 + HKDF-SHA256 key wrapping
Registry	FastAPI, SQLite (named Docker volume), Caddy (TLS, local CA), Docker Compose
Clinic portal	Flask, Jinja2, gunicorn, SQLite (WAL), server-rendered HTML + one self-hosted JS file
Patient data security	AES-256-GCM blob store, Argon2id passwords, TOTP two-step sign-in, CSRF tokens, strict CSP, hash-chained audit log
Device	Raspberry Pi 4 (4 GB) with the official 7-inch display, Raspberry Pi OS (Debian 13), Caddy HTTPS, nftables firewall, systemd units, Tailscale Funnel for remote access
Kiosk	labwc (Wayland), Chromium in kiosk mode, locked-down user, built-in touch keyboard
Interface	IBM Plex Sans and Plex Mono (self-hosted), token-based design system, canvas 3D brain renderer
Testing	pytest (179 automated tests), live end-to-end registry check, scripted browser walk-through (this report's figures)

Table 1: Technology stack.

4 Data and leakage control

4.1 Two-dimensional training data

Training uses BRISC 2025, figshare (Cheng 2015) and the Nickparvar collection; BDNeuro-MRI is an external test set. Every image is compared with every other using a flip-aware normalised cross-correlation of 64×64 thumbnails. A copy threshold of 0.98 sits in an empty gap of the similarity distribution; a same-patient threshold of 0.94 was calibrated on figshare's true patient identifiers ($\leq$1% false links). Splits are grouped so that copies and same-patient images never straddle train and test.

Finding	Count
BRISC official test images that are copies of training images	566 / 1,000 (462 mirrored)
Nickparvar images byte-identical to BRISC	4,775 / 7,200
figshare images that are near-duplicates of BRISC	2,873 / 3,064
BDNeuro images that are copies or mirrors of BRISC	3,768 / 5,941

Table 2: Leakage found in public brain-tumour MRI collections.

4.2 Three-dimensional patient studies

After the radiologist's feedback the system was evaluated on complete studies. Only the T1 post-contrast volume and the tumour mask are kept per patient.

Dataset	Class	Patients	Role and licence
BraTS 2023 adult glioma	glioma	300	development; research use (Synapse)
BraTS 2023 meningioma	meningioma	300	development; CC BY-NC
NHNN pituitary MRI	pituitary	60	development, held-out part tested; CC BY
IXI healthy volunteers	no tumour	581	30% calibration, 70% test, never trained on; CC BY-SA 3.0
BraTS-Africa	glioma	146	external test (pending); CC BY 4.0

Table 3: 3D datasets. Splits are by patient; a cross-dataset similarity check excluded 29 IXI volunteers whose images were already in the 2D training data.

Large archives were not downloaded whole: the pituitary set (47 GB) and BraTS were read with HTTP range requests, so only the needed files (about 1.2 GB for 600 BraTS patients) were transferred.

5 Model

5.1 Architecture

A ResNet-50 backbone at output stride 8 produces a 28×28 feature map A. An FPN segmentation head predicts the lesion probability p. The classifier uses *lesion-attention pooling*:

$$z_c = b_c + \sum_{ij} a_{ij}\, W_c^{\top} A_{ij}, \qquad a_{ij} = \frac{p_{ij} + \varepsilon}{\sum_{kl}(p_{kl} + \varepsilon)}, \tag{1}$$

where p is the detached segmentation probability pooled to the feature grid and $\varepsilon = 0.005$. With no lesion the weights are uniform (ordinary average pooling). The exact class-evidence map $S_c(ij) = a_{ij} W_c^{\top} A_{ij}$ sums to the logit (HiResCAM form), so the Grad-CAM shown to the radiologist is an exact decomposition of the decision, not an approximation, and it is lesion-focused by construction.

5.2 Two-dimensional results (v1, deployed INT8 graph)

Metric	v0	v1 (deployed)	v1, 3 seeds
Test macro-F1 [95% CI]	0.935	**0.944** [0.926, 0.959]	0.938 ± 0.007
External BDNeuro macro-F1	0.729	**0.836**	0.827 ± 0.012
Grad-CAM evidence inside tumour	0.150	**0.728** (98% of ceiling)	0.713 ± 0.018
Pointing game	0.525	**0.992**	0.981 ± 0.019
95% prediction-set coverage	0.867	**0.911** (target 0.93)	0.917 ± 0.006

Table 4: Single-image results on the leakage-free test set (799 images).

5.3 Whole-study evaluation and model v2

For a study, 36 slices are taken in three planes (12 axial, 12 coronal, 12 sagittal), each is analysed, and the patient result is the mean of the five slices most suggestive of tumour (*top-k*, chosen on calibration patients only). Targets were fixed before testing.

Held-out patients	Target	v1	v2 (`mri/0.3.0`)
Tumour-type macro-F1 (159 patients)	≥ 0.85	0.71	**0.98**
Slice tumour AUROC (5,724 slices)	≥ 0.90	0.72	**0.95**
Healthy false-alarm rate (IXI, 380)	$\leq 10\%$	23%	**5%**
Glioma accuracy (76)		54%	100%
Meningioma accuracy (68)		87%	91%
Pituitary accuracy (15)		53%	100%
Healthy volunteers correct (380)		77%	95%

Table 5: Patient-level results. v2 keeps v1's architecture and preprocessing and adds slices from 3D training patients (tumour and tumour-free slices of the same scans).

The test patients come from the same datasets as the training patients; the external whole-study test (BraTS-Africa) and two further training seeds remain to be done.

6 Explainability and uncertainty

- **Grad-CAM (exact).** Shown on the key slices with the lesion outline from the segmentation head. The INT8 decomposition error is about 10^{-7}.
- **Calibration.** Temperature scaling fitted on a dedicated calibration split of the deployed INT8 graph (expected calibration error 0.049 to 0.030 on test).
- **95% prediction set.** Class-conditional conformal prediction; a set with more than one class marks the study for expert review.
- **Out-of-distribution detection.** A multi-layer Mahalanobis score with a warning threshold (95% of validation) and a reject threshold (99.5%). Rejected inputs show no diagnosis.
- **Review flags.** Thresholds calibrated on validation data, for example the share of Grad-CAM evidence inside the predicted lesion.

A local 1.7B language model was evaluated as a report assistant. About 4% of its verified answers still contained a clear error, mostly misreading statistics, so it was removed from the portal and remains only as a research experiment.

7 Edge optimisation and deployment

The device runs no PyTorch: preprocessing is shared code (NumPy and OpenCV) with a fingerprint check against the bundle, and inference is ONNX Runtime on the CPU.

Measure	Raspberry Pi	PC (i7)
Single image, full analysis (median / p95)	238 / 250 ms	97 ms
Whole study (36 slices + key-slice overlays)	43–51 s (Pi 4 kiosk)	3–6 s
Peak memory, large 0.5 mm study	2.5 GB	–
Pi vs PC agreement (40 scans)	40/40, CAM $r \geq 0.997$	–
Model size (INT8)	24.7 MB (FP32 96.6 MB)	

Table 6: Latency and parity. Only one whole-study analysis runs at a time to bound memory.

The installer (`edge/install_pi.sh`) is idempotent and sets up the Python environment with pinned versions, Caddy HTTPS, the firewall, the portal key and database, systemd services, model sync, and optionally the kiosk.

8 Model security and storage

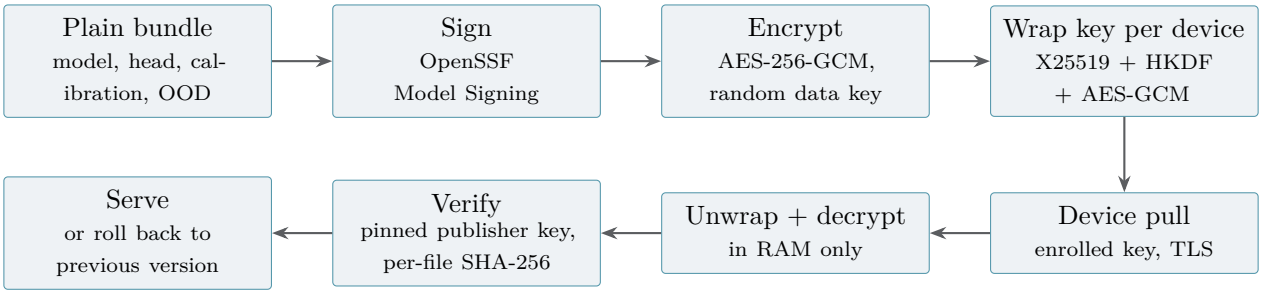


Figure 3: Model distribution. The signing key never leaves the training PC; the server sees only public keys and ciphertext; the device stores only ciphertext.

- **Provenance.** Bundles are signed with OpenSSF Model Signing. The device pins the publisher's public key at enrolment and never fetches it from the server, so a compromised server cannot substitute a model.
- **Confidentiality.** Envelope encryption: the bundle is encrypted once with AES-256-GCM; the data key is wrapped separately for every enrolled device with X25519 ECDH, HKDF-SHA256 and AES-GCM. The model name and version are bound as associated data, so a key or ciphertext cannot be replayed against another version.
- **Enrolment and revocation.** One-time enrolment tokens; per-device API keys; revoked devices receive 403.
- **Storage on the device.** Only the encrypted blob and its manifest are stored (`/var/lib/neuroedge/models`); the stored blob measures 8.000 bits per byte of entropy. Decryption happens in memory at start-up, and if the newest version fails verification the previous one is used.
- **Live verification.** `server/e2e_check.py` runs 16 checks against the live registry (verified TLS, signature and decryption, byte-identical files, ciphertext-only storage, 401 on forged keys, a single flipped ciphertext bit detected, another device unable to decrypt, 403 after revocation). All 16 pass for `mri/0.3.0`.

9 Portal security and patient-data storage

Concern	Implementation
Transport	HTTPS through Caddy with a local certificate authority; the Flask app listens on loopback only; nftables admits only SSH, HTTP(S) and mDNS
Scans, overlays, reports	Encrypted blob store: AES-256-GCM, a 12-byte nonce per file, the case and file kind as associated data so files cannot be swapped between cases; key in `/etc/neuroedge/portal.key` (root and service only)
Records	SQLite in WAL mode; the database stores SHA-256 hashes of every scan and signed report snapshot
Accounts	Argon2id password hashing, account lock-out, optional mandatory two-step sign-in (TOTP with recovery codes) for staff
Sessions and forms	HttpOnly, SameSite=Strict, `__Host-` cookies; CSRF token on every form; idle and absolute session timeouts
Browser hardening	Content Security Policy allows only same-origin scripts and styles, no inline code; frame and referrer protection
Access control	Roles (patient, radiologist, coordinator) and object-level checks; patients never receive AI overlays or probabilities
Accountability	Hash-chained, tamper-evident audit log of every view, signature and change, verifiable from the command line
Privacy	Consent records, full data export for the patient, deletion requests, e-mails with no clinical content

Table 7: Security controls in the clinic portal.

10 The clinical workflow, step by step

The figures in this section were produced by a scripted walk-through of the running system (`docs/report/capture.py`): a fresh demo database, a real whole-study MRI upload analysed by model v2, and the e-mails exactly as the portal generated them.

10.1 Device home and sign-in

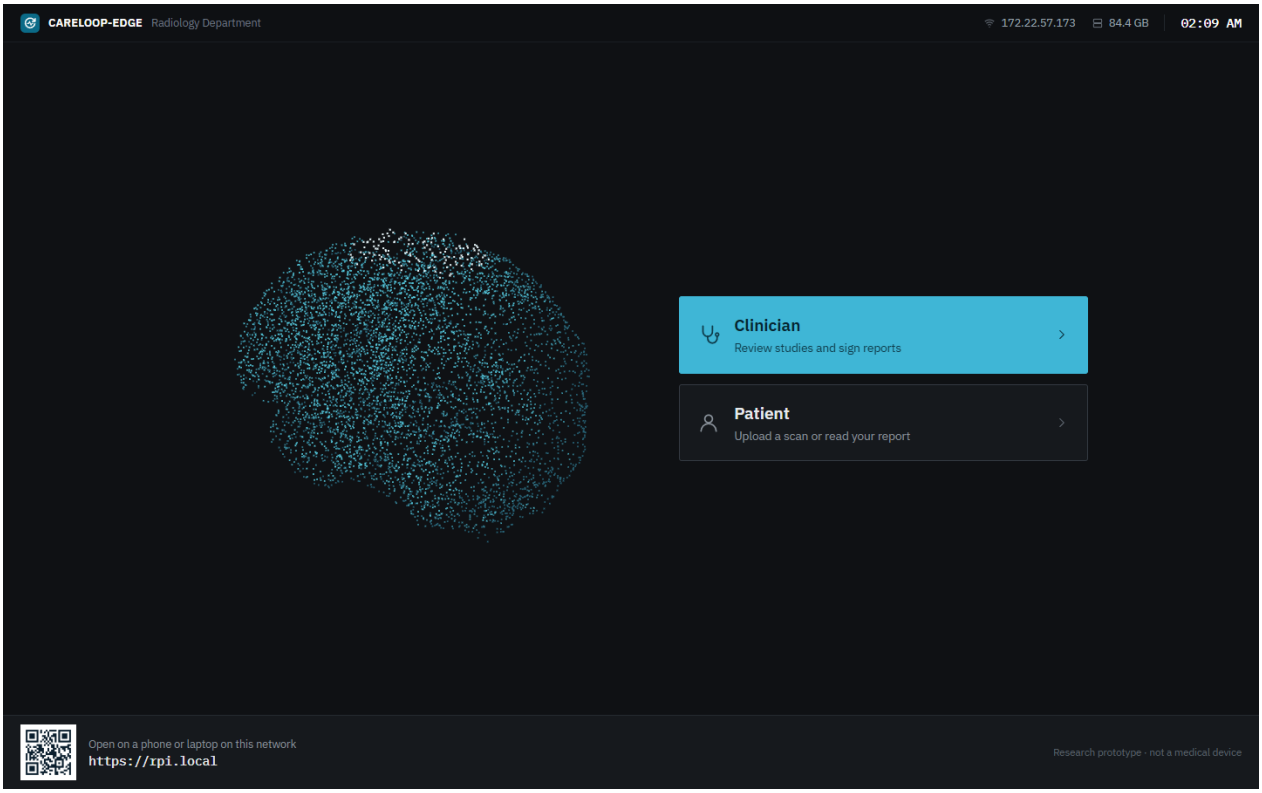


Figure 4: Device home. The portal's front page: a rotating brain surface reconstructed from a real MRI study, a status bar (network address, storage, clock) and the two entry points. No patient data appears here.



Figure 5: The same page on the official 7-inch touch display (800×480).

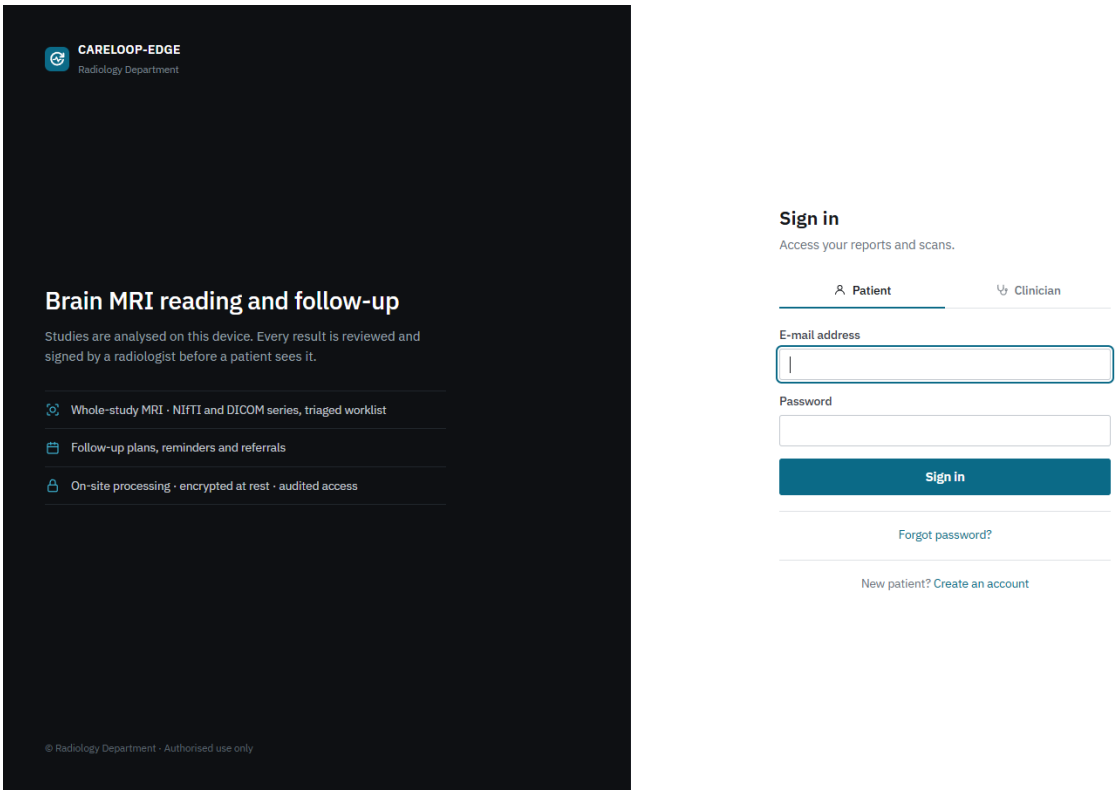


Figure 6: Sign-in, with separate patient and clinician tabs.

10.2 Patient: registration and upload

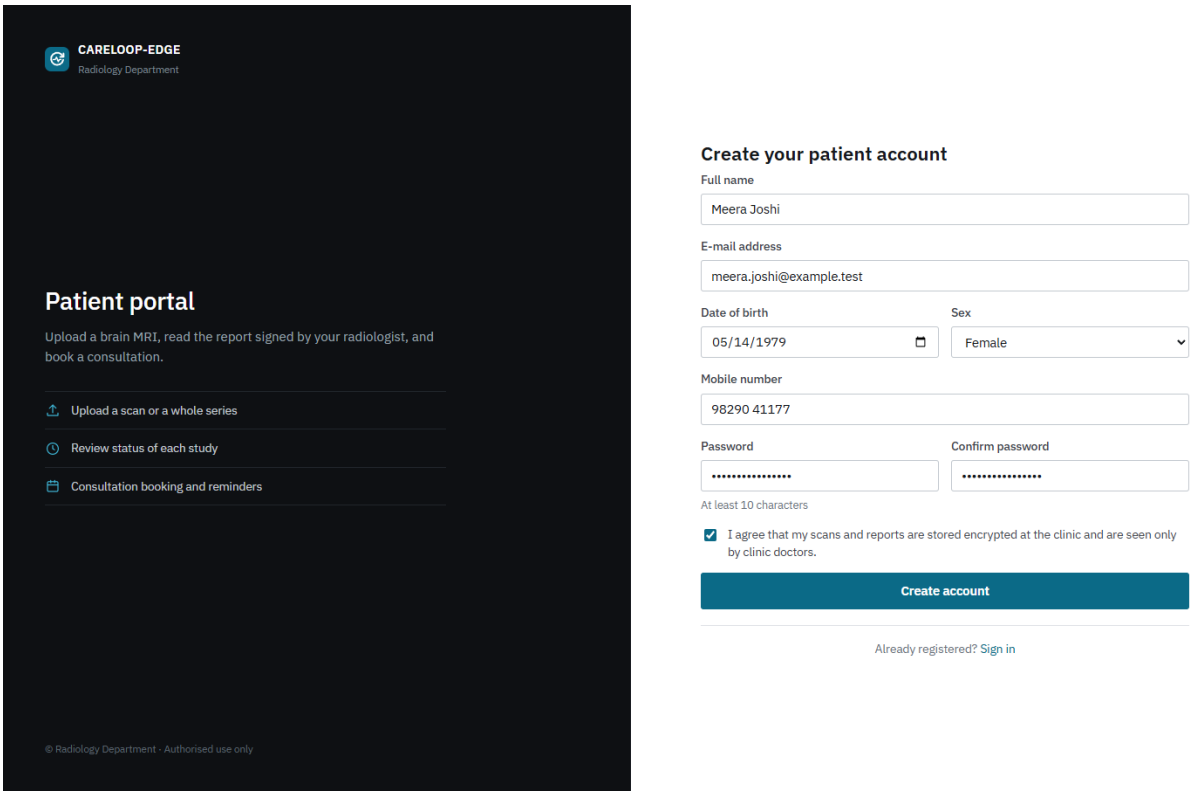


Figure 7: Patient registration with explicit consent for storage and review.

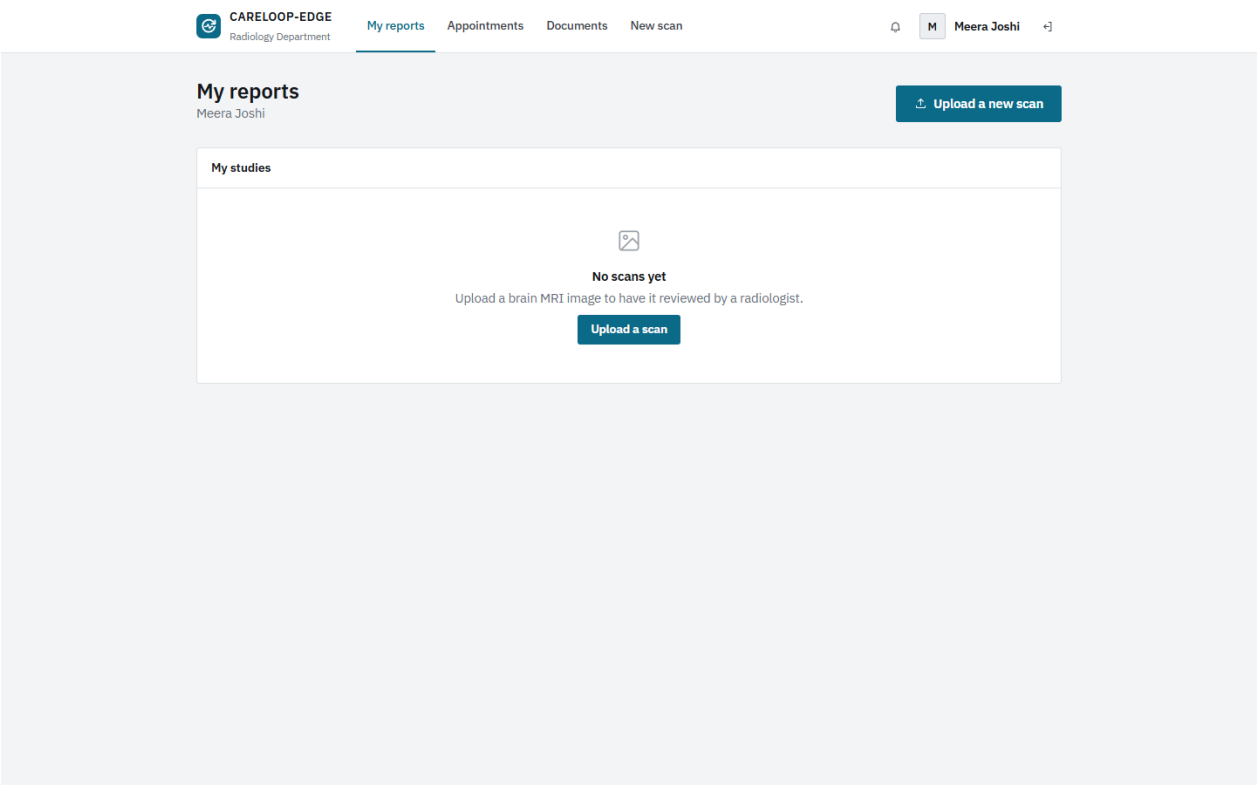


Figure 8: A new patient's dashboard.

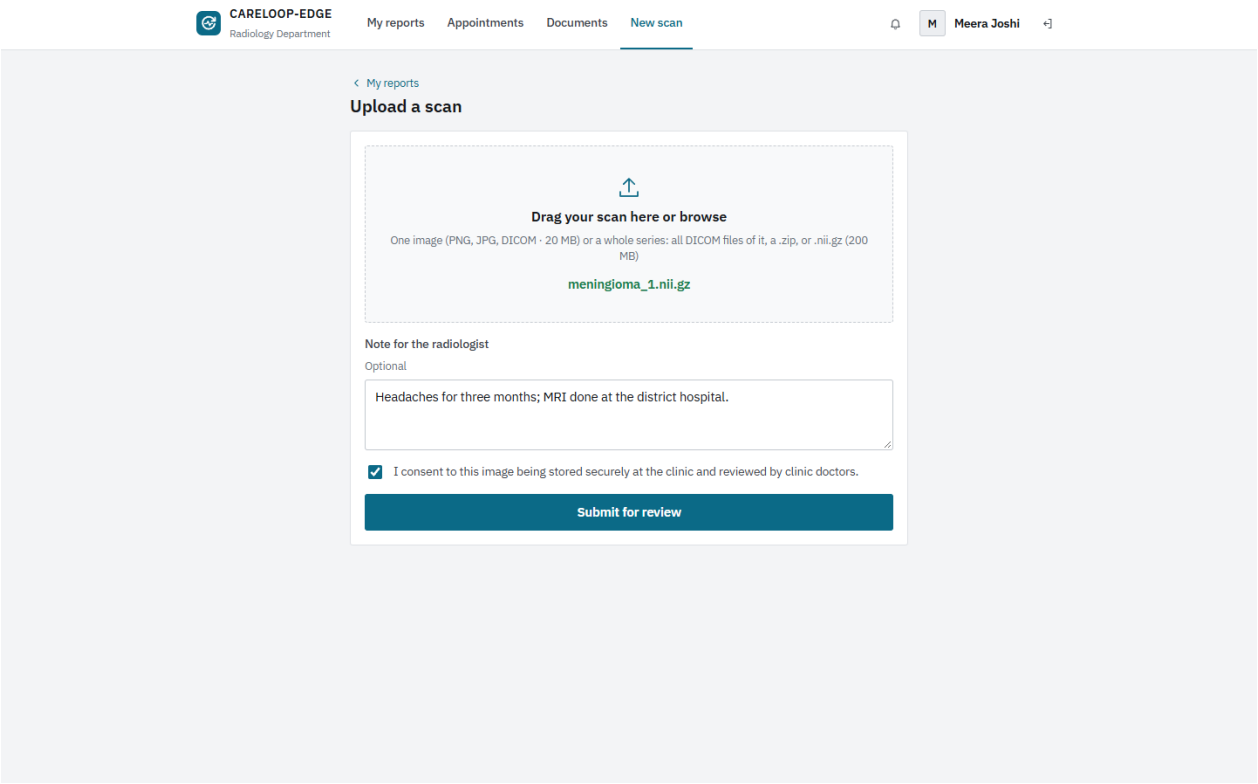


Figure 9: Uploading a whole MRI study (a NIfTI volume) with a note for the radiologist. Single images, ZIP archives and multi-file DICOM series are also accepted.

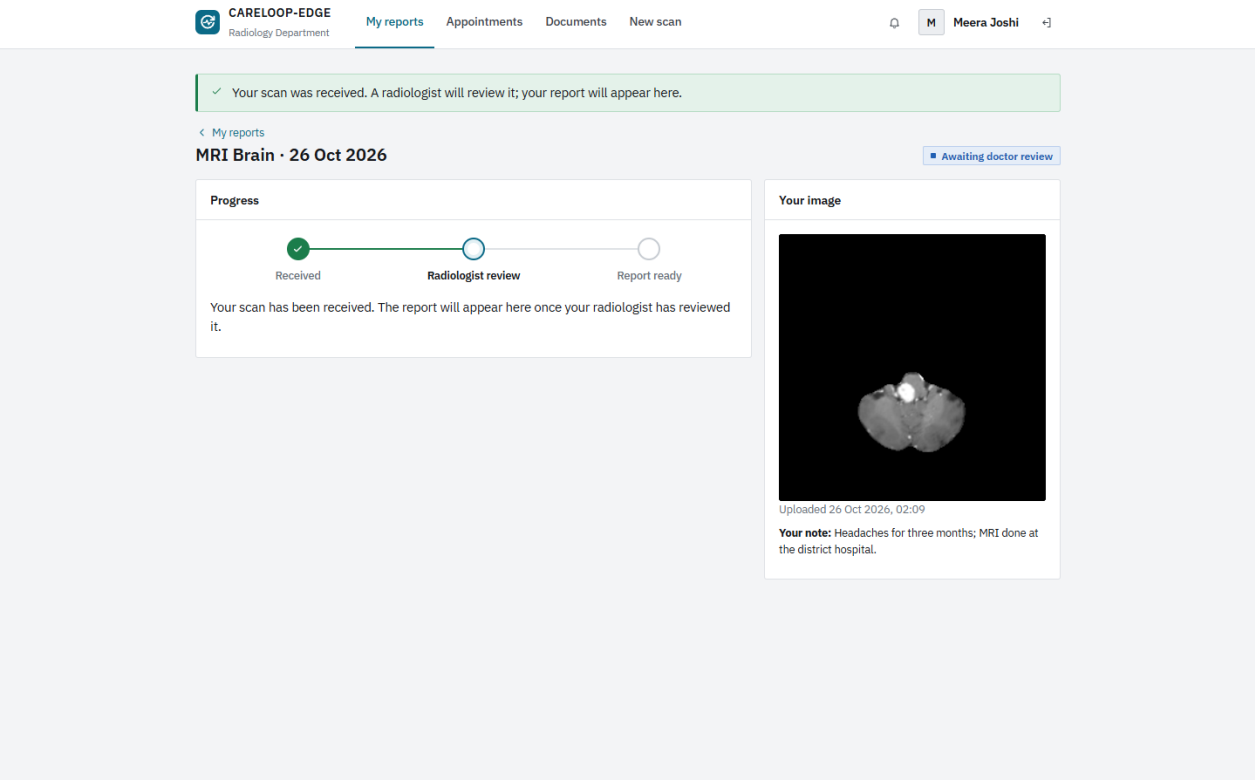


Figure 10: After upload: the study is received and the progress tracker shows that it is waiting for the radiologist. The patient sees no AI result.

10.3 Radiologist: worklist, review and signature

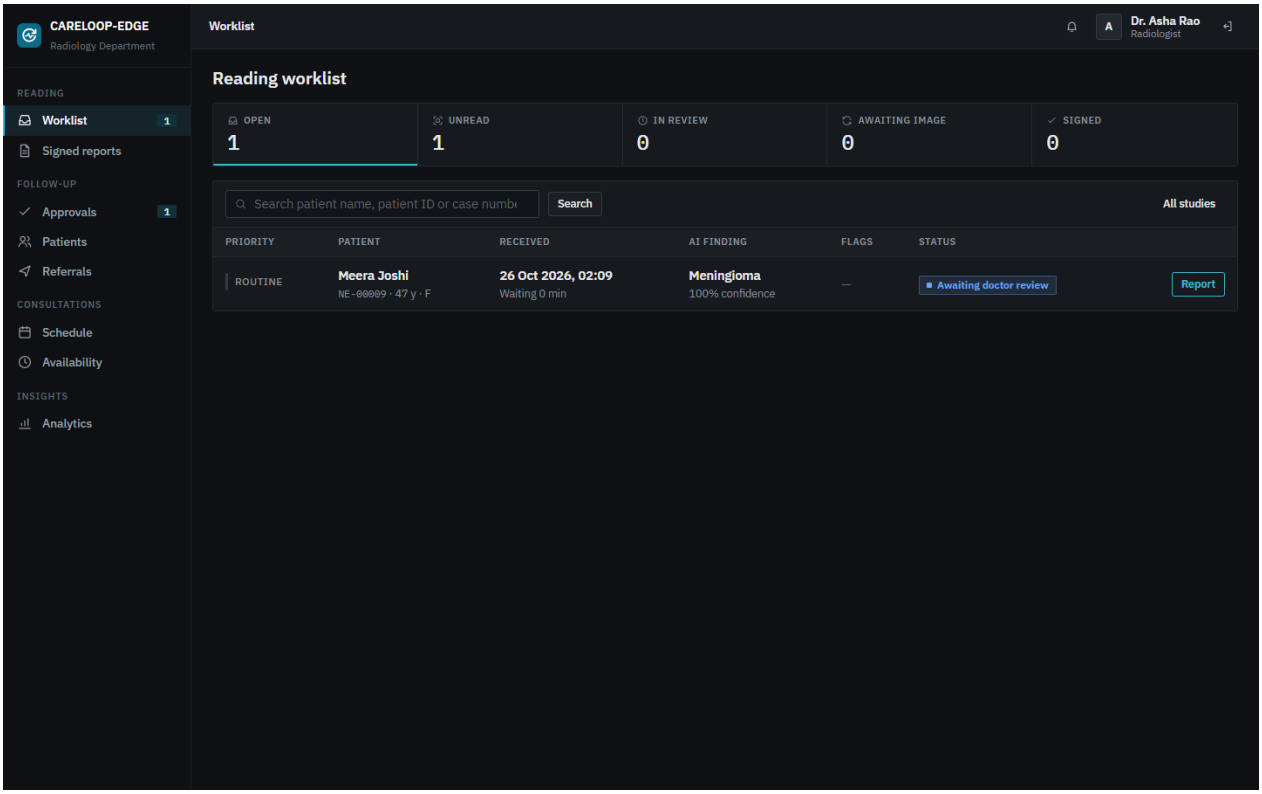


Figure 11: The reading worklist, prioritised by the AI's review flags, with waiting times and status.

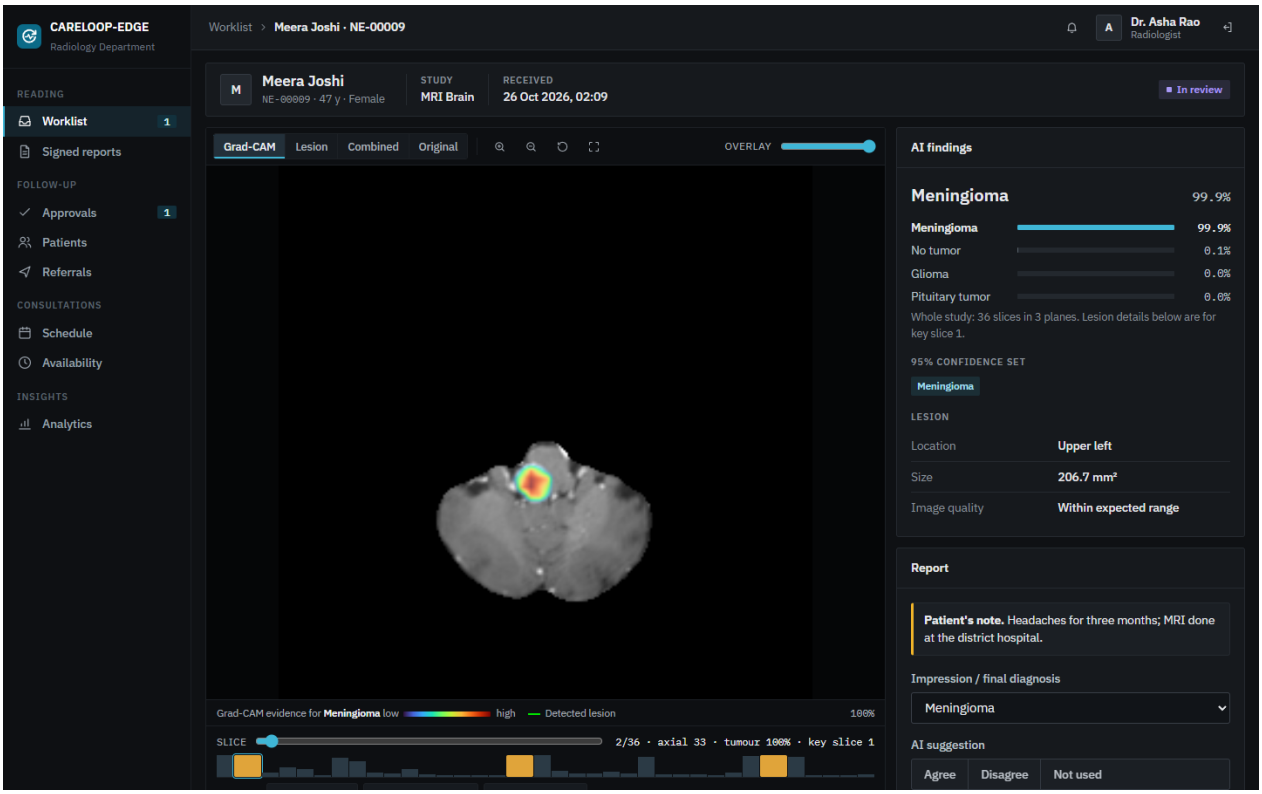


Figure 12: Case review. The viewer shows Grad-CAM on key slice 1; the slice strip below shows the tumour likelihood of all 36 slices (key slices in amber). The findings panel gives the study-level probabilities, the 95% prediction set and the lesion measurements.

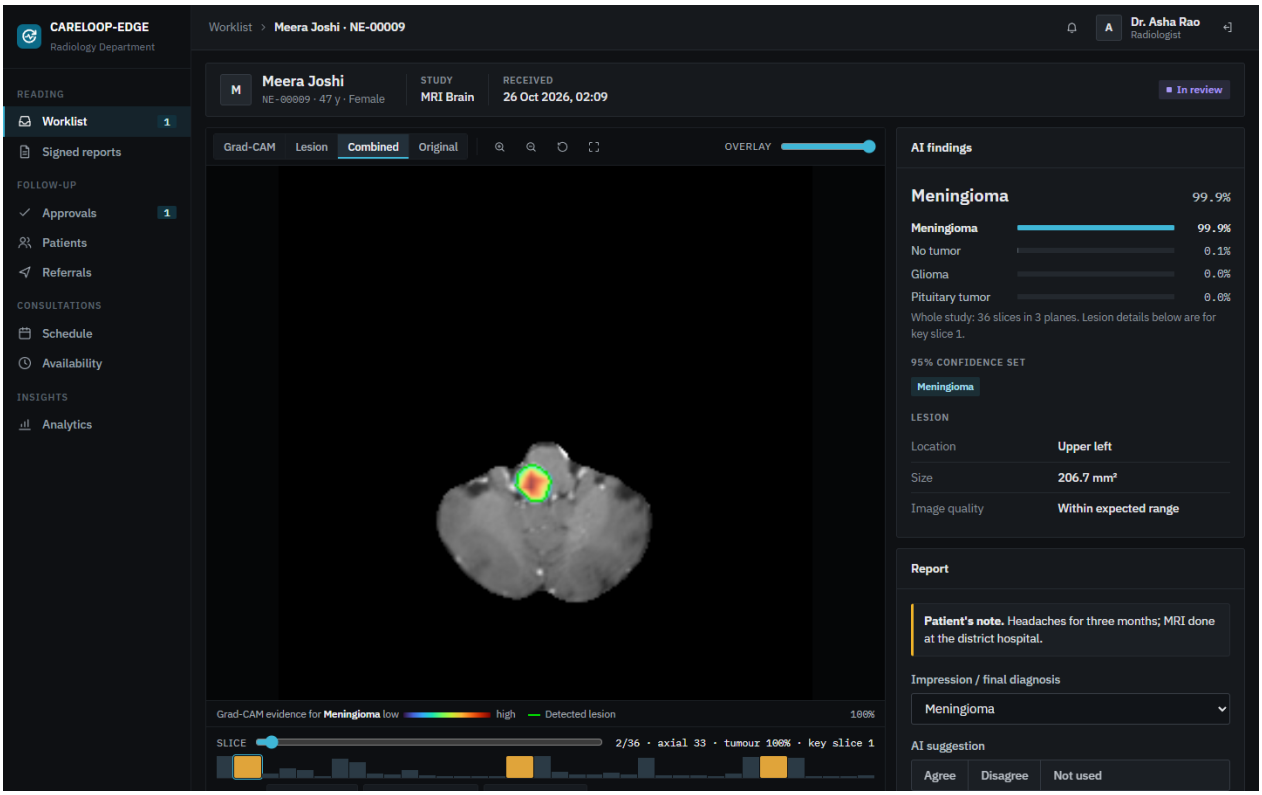


Figure 13: Combined view: Grad-CAM with the lesion outline from the segmentation head.

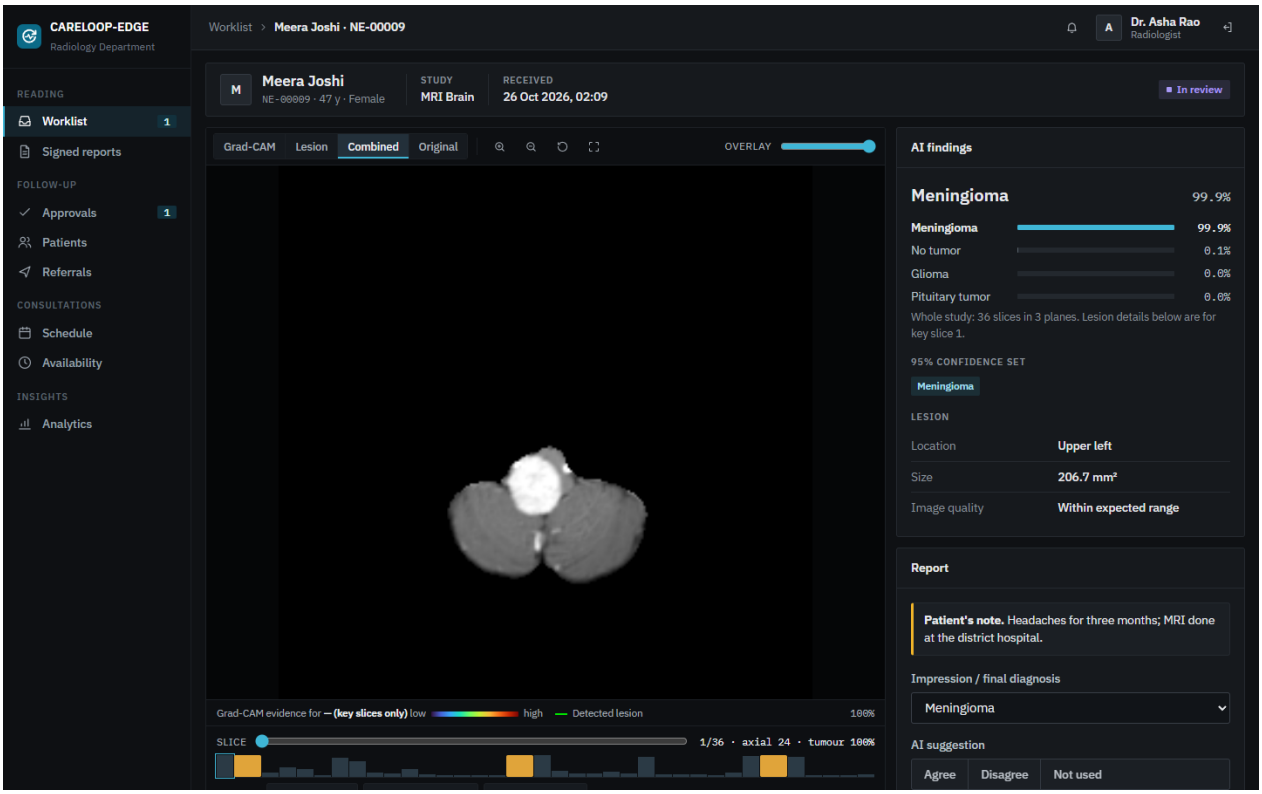


Figure 14: Scrolling through the study: a non-key slice is shown without overlays, as the radiologist reads the whole series.

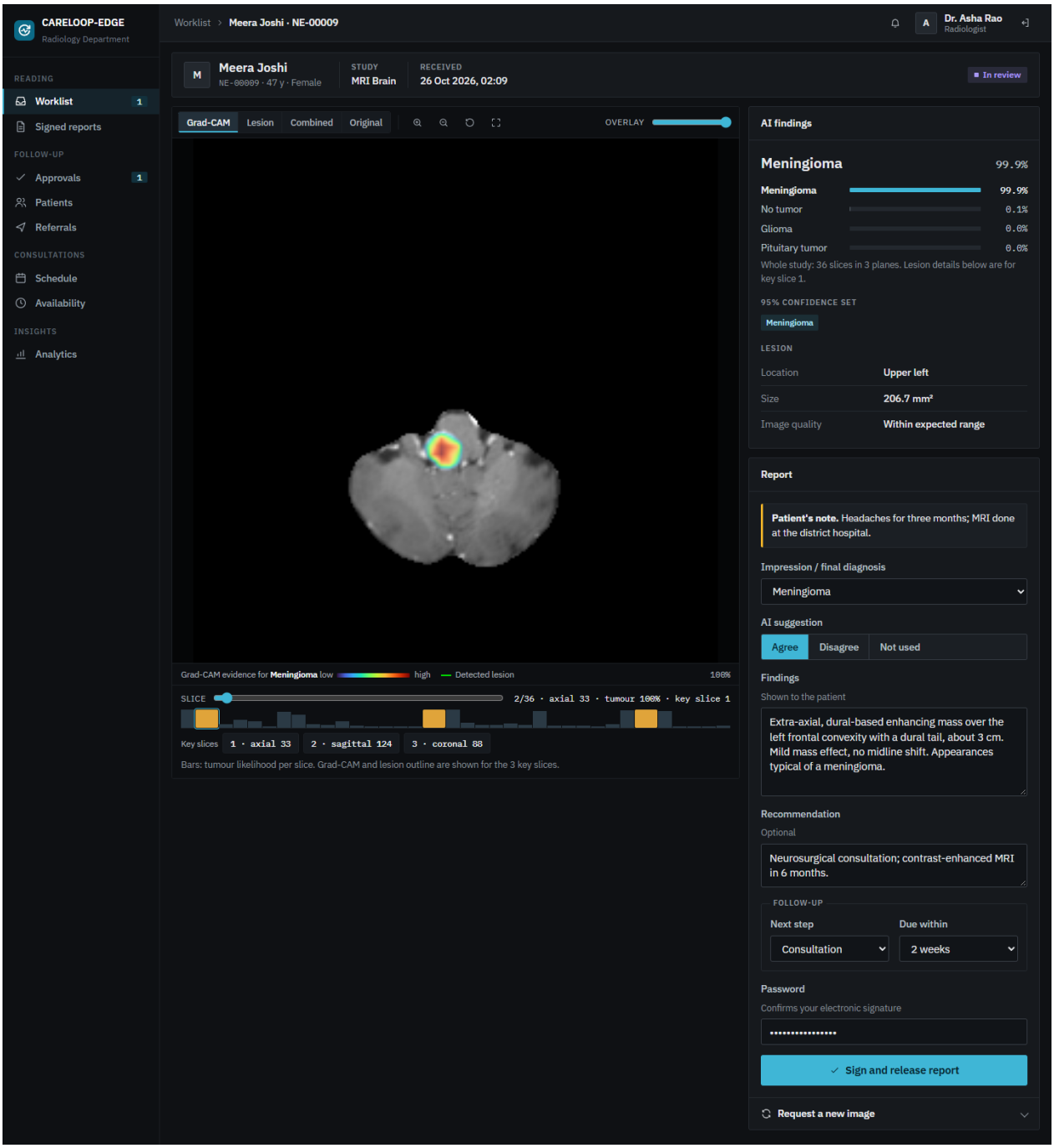


Figure 15: Signing the report. The radiologist states agreement with the AI, writes findings and a recommendation, and chooses the follow-up step (here a consultation within two weeks). The signature requires the password again.

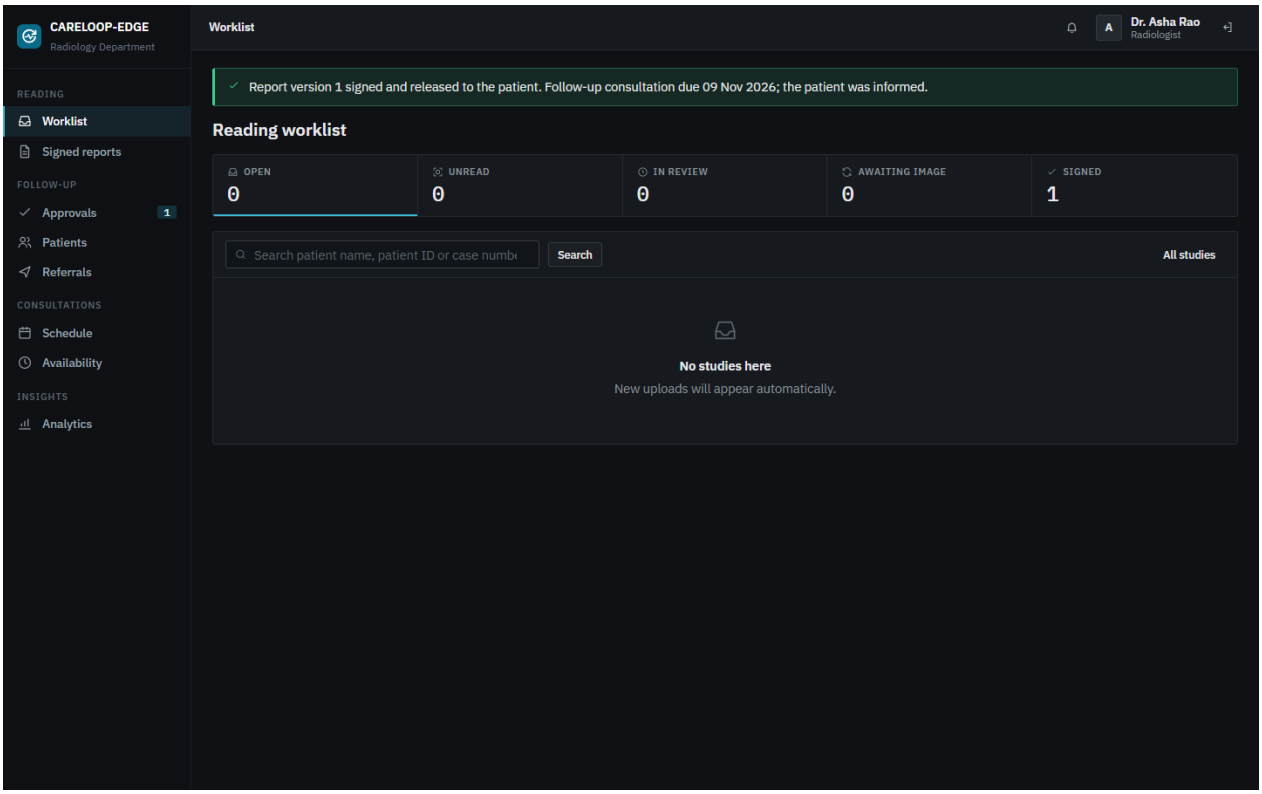


Figure 16: The signed report is released to the patient and the follow-up plan is created.

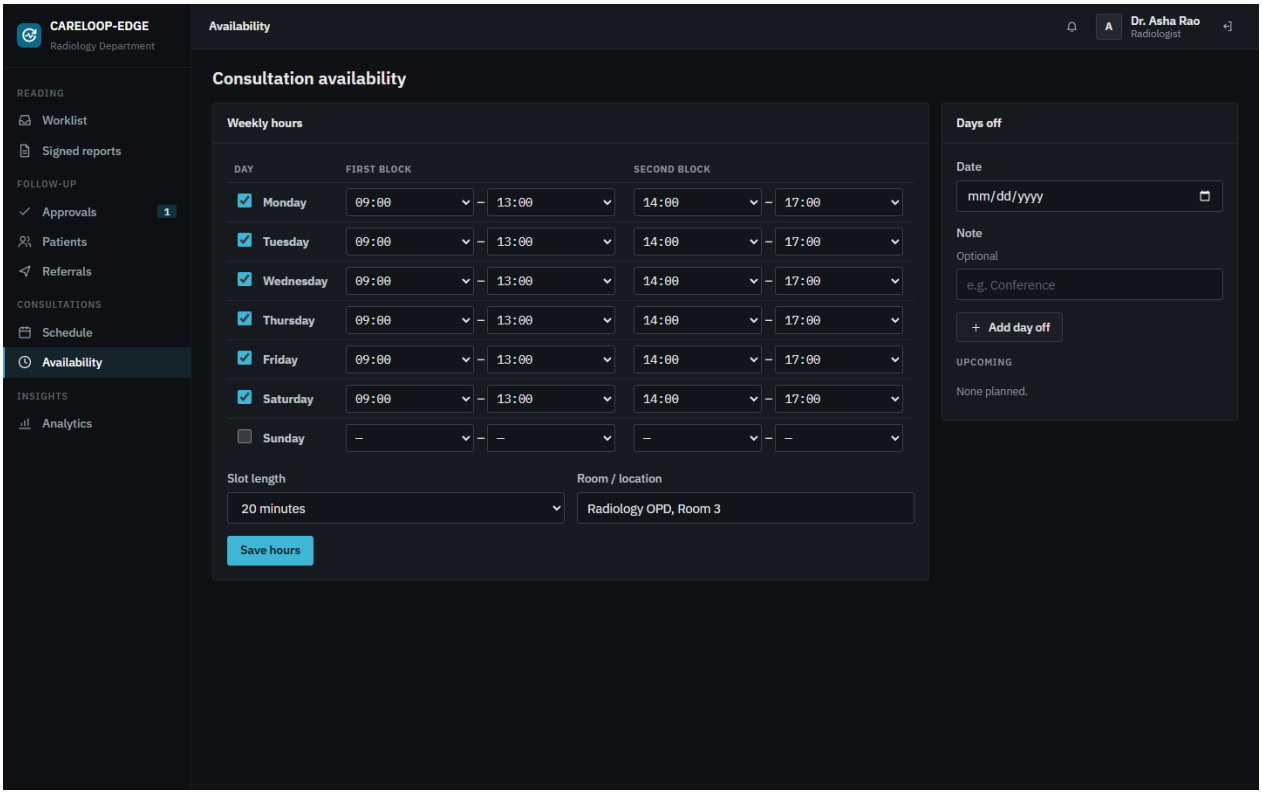


Figure 17: Consultation hours and days off, which define the slots patients can book.

10.4 Patient: report, next step and booking

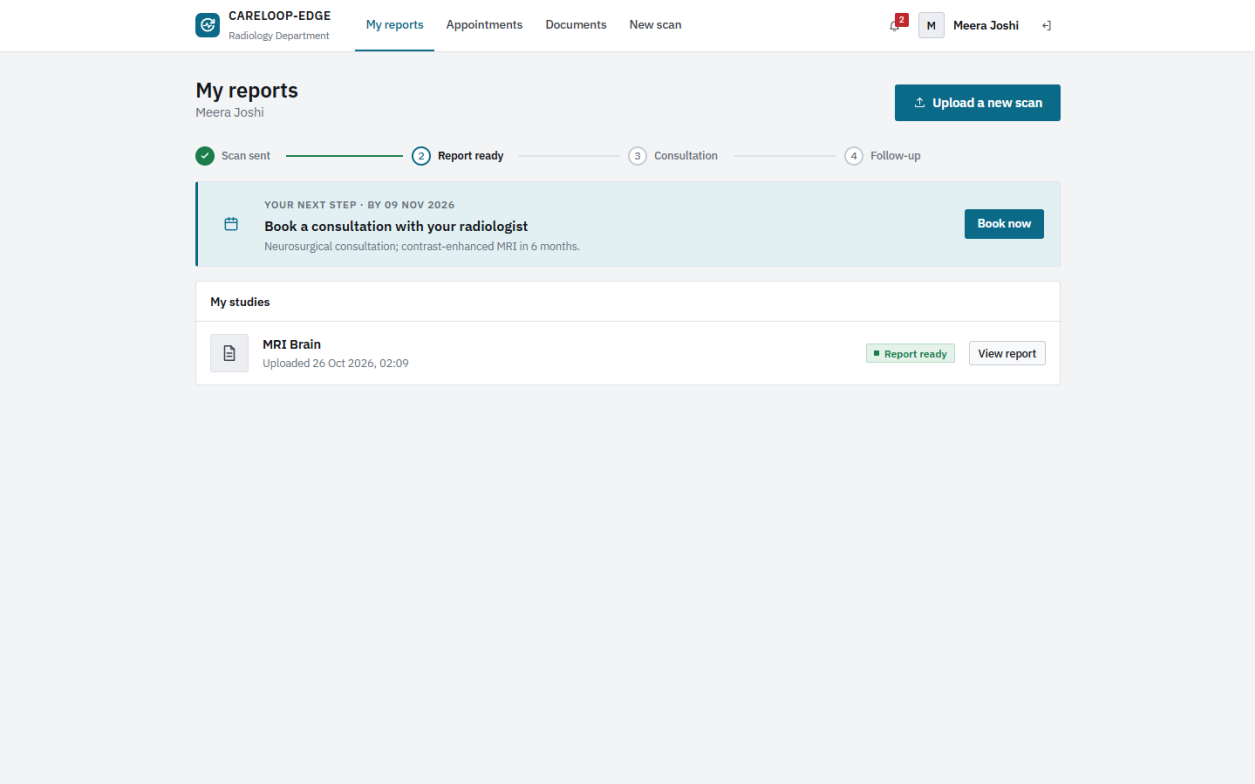


Figure 18: The patient's dashboard after signing: the care-progress tracker and the next step requested by the radiologist.

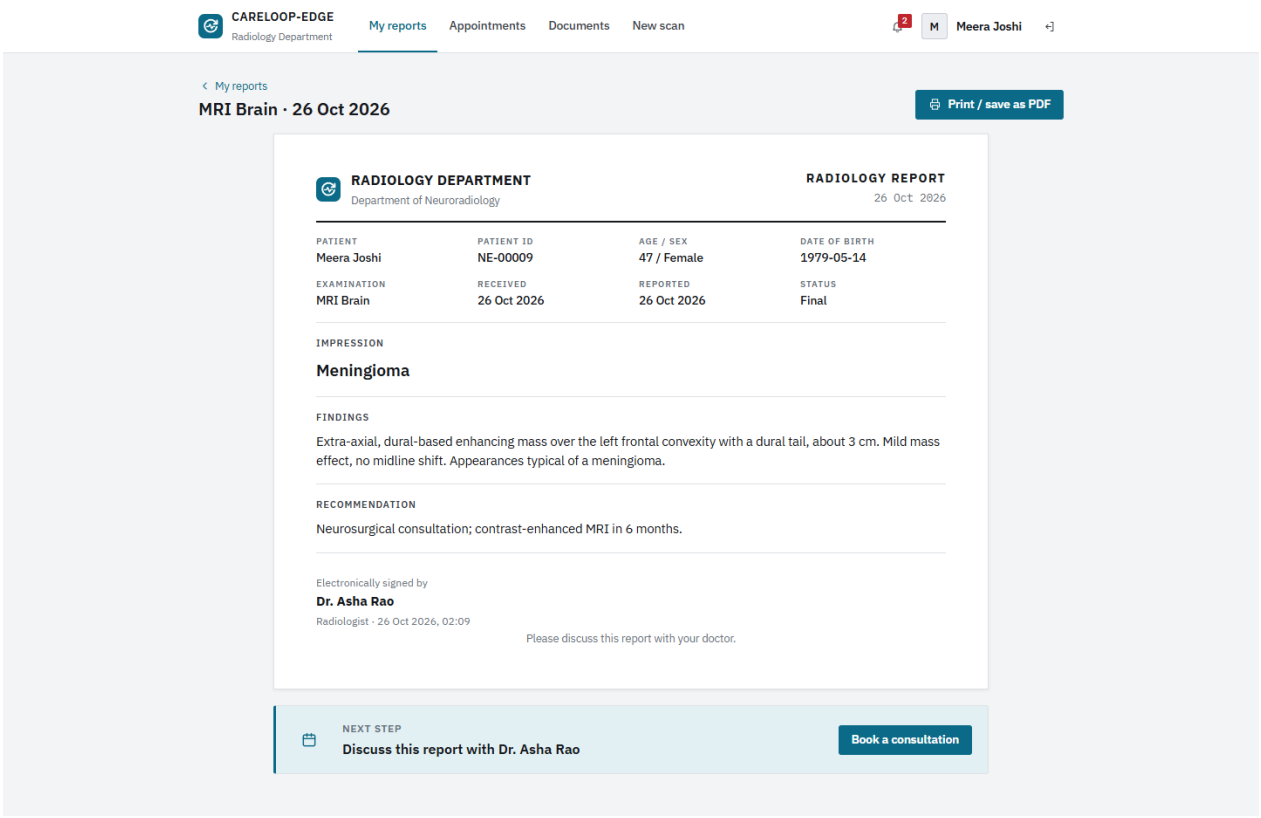


Figure 19: The signed radiology report, as the patient sees it (printable). It contains the radiologist's diagnosis and findings only, never AI output.

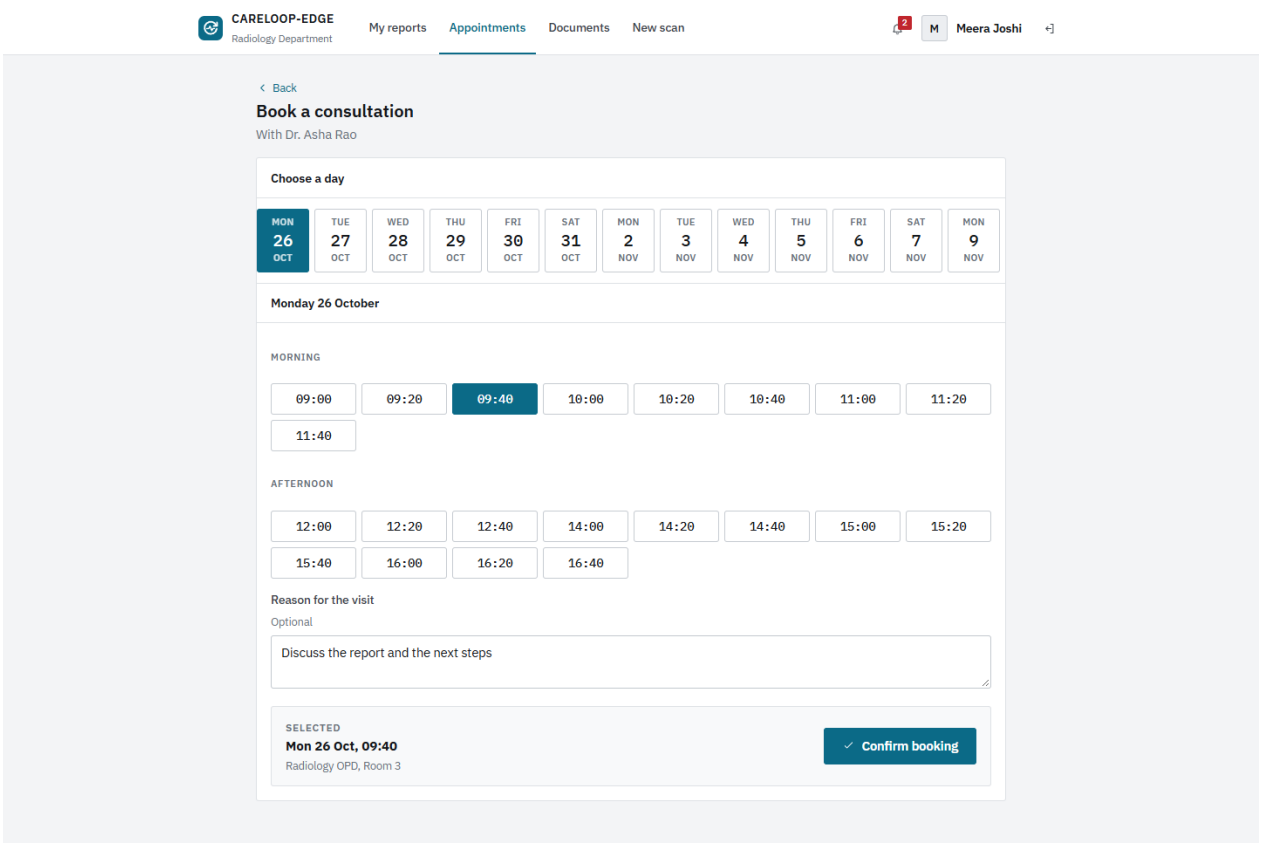


Figure 20: Booking a consultation from the radiologist's free slots.

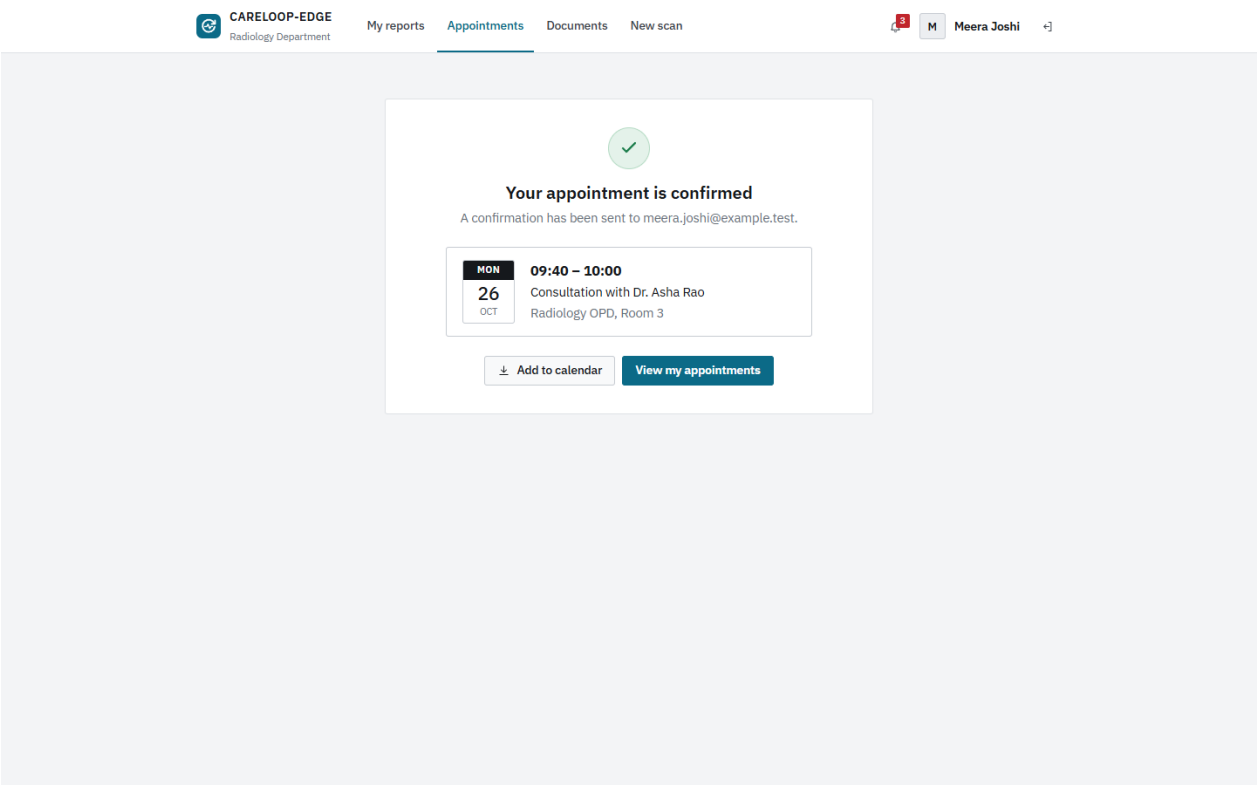


Figure 21: Booking confirmation with a calendar invitation.

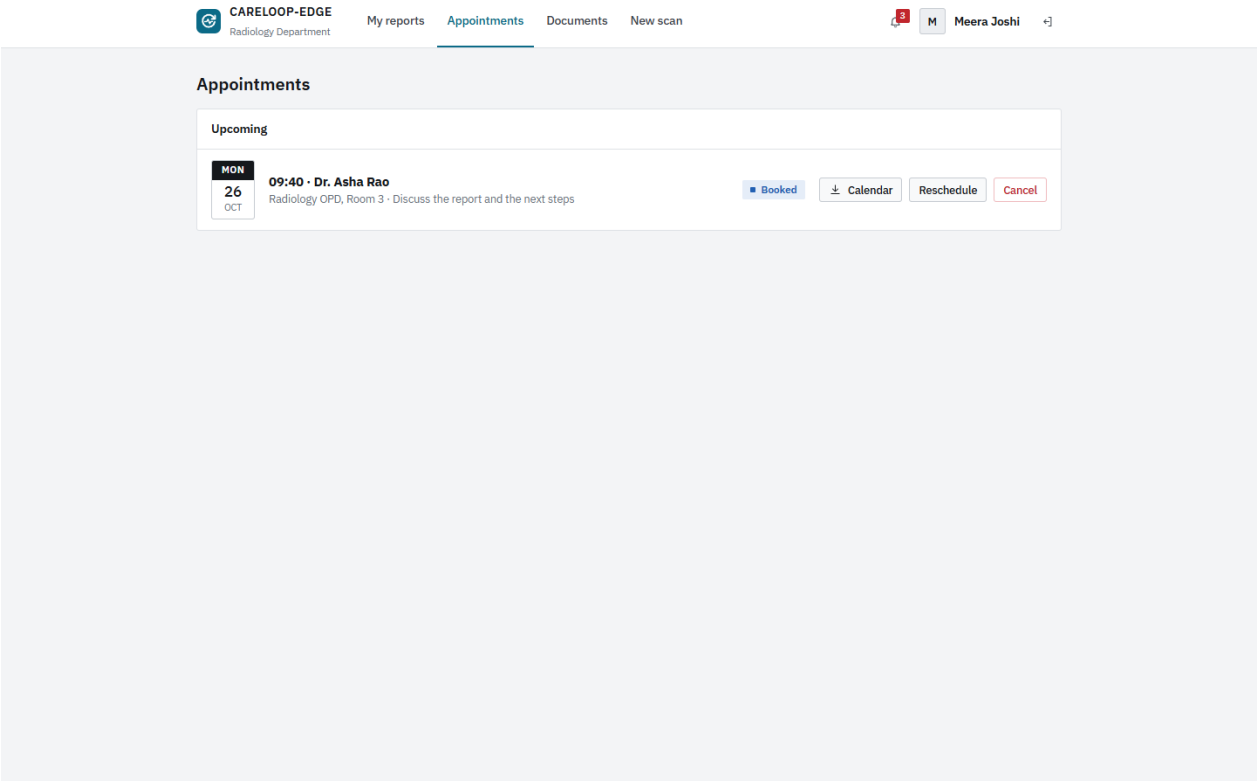
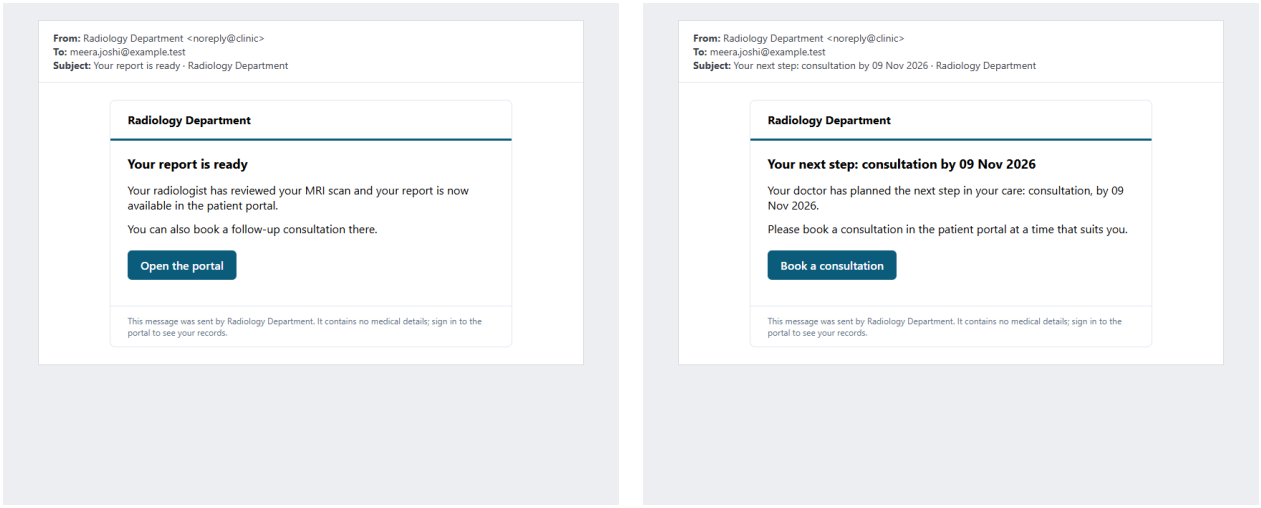


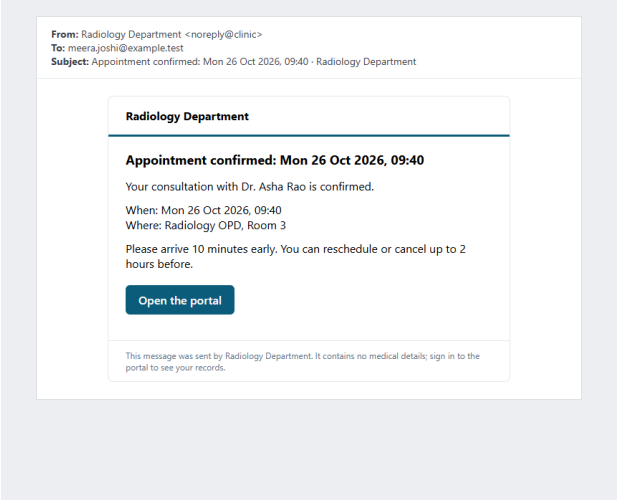
Figure 22: The patient's appointments, with options to reschedule or cancel.

10.5 E-mails sent along the way

E-mails are sent through the clinic's SMTP server (here saved as files in demo mode). They carry no diagnosis or other medical details; the patient signs in to read the report.



(a) Report ready. **(b)** Next step requested.



(c) Appointment confirmed.

Figure 23: The three e-mails generated in this walk-through.

10.6 Radiologist: schedule and analytics

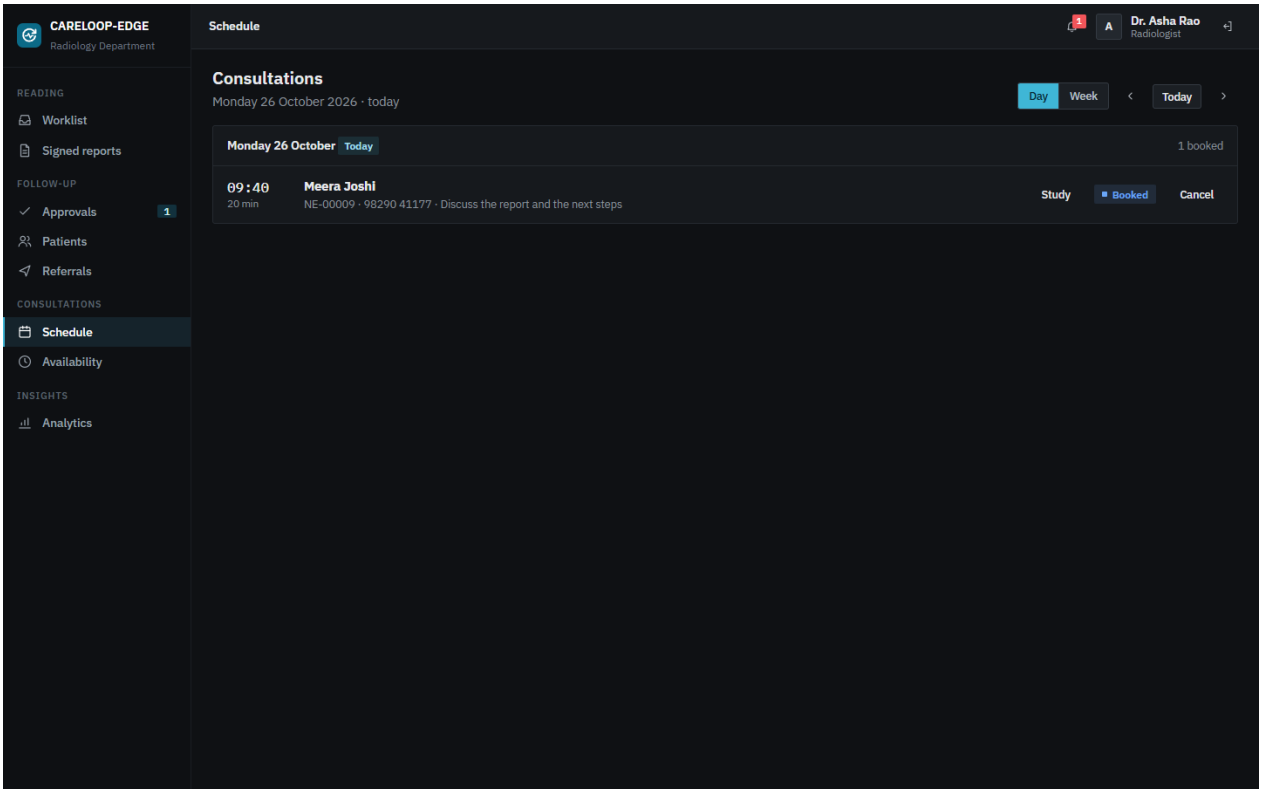


Figure 24: The radiologist's consultation schedule with the patient's booking.

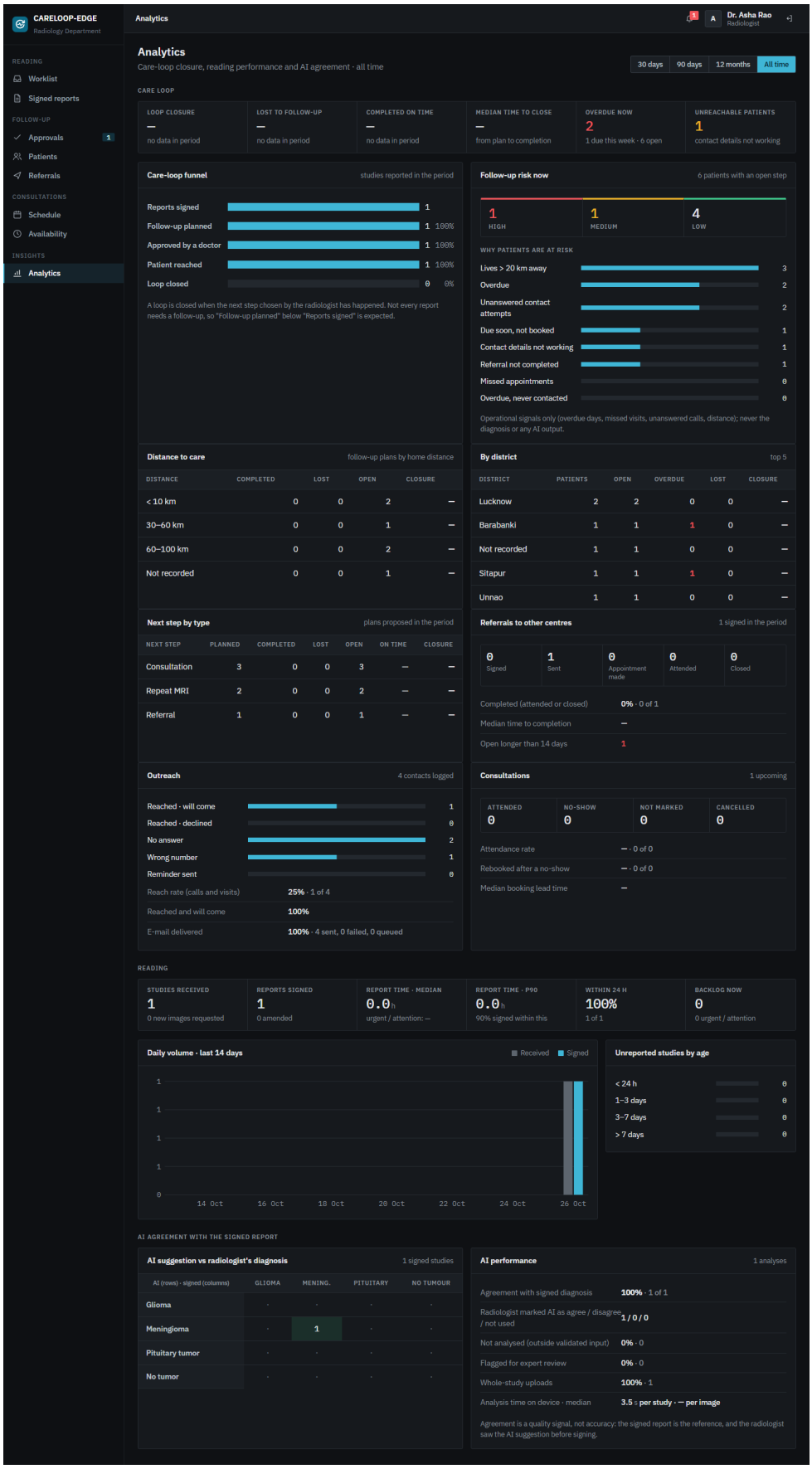


Figure 25: Analytics. The care-loop section (closure and loss-to-follow-up rates, a funnel from signed report to closed loop, risk bands with their reasons, distance to care, districts, next steps, referrals, outreach and consultations), followed by reading performance and AI agreement with the signed diagnosis. Every rate shows its denominator.

10.7 Care coordinator: follow-up and administration

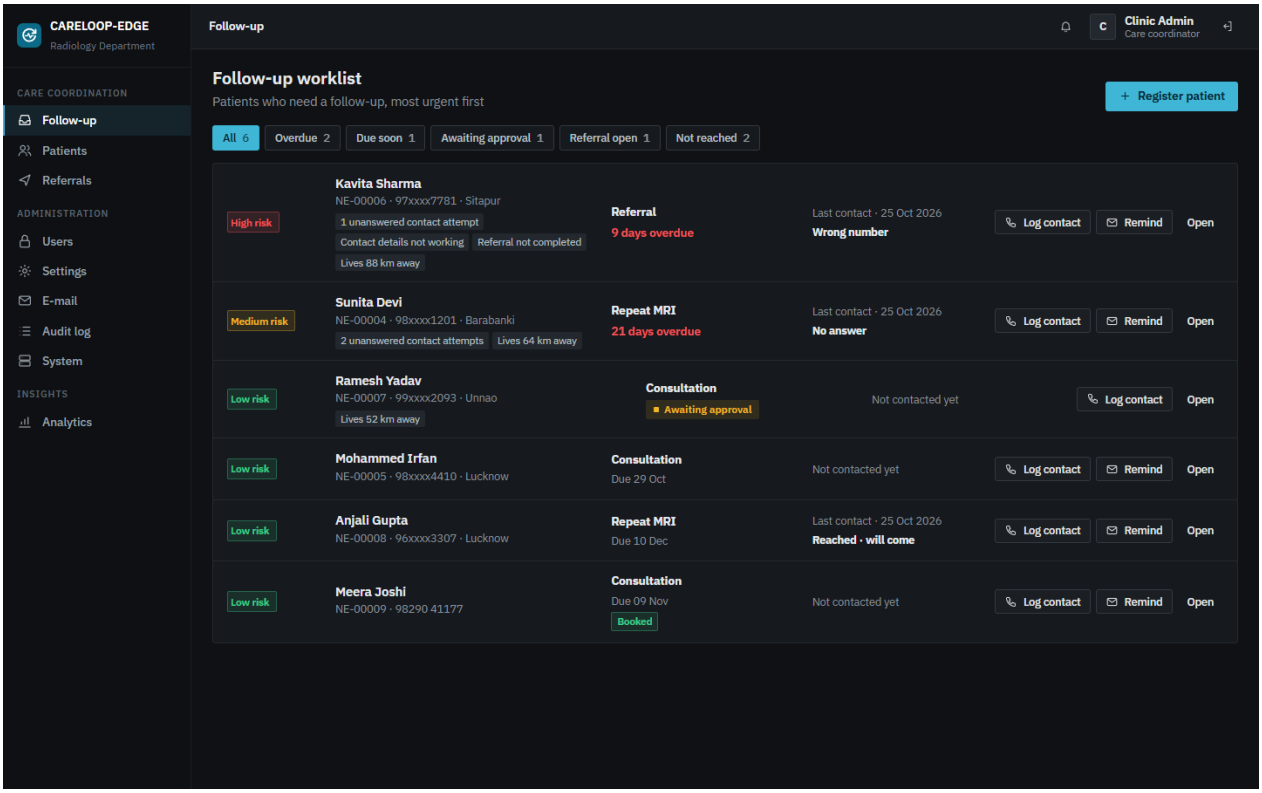


Figure 26: The follow-up worklist, ranked by an explainable risk score (days overdue, missed visits, unanswered calls, contact details, open referral, distance). The score never uses the diagnosis or any AI output.

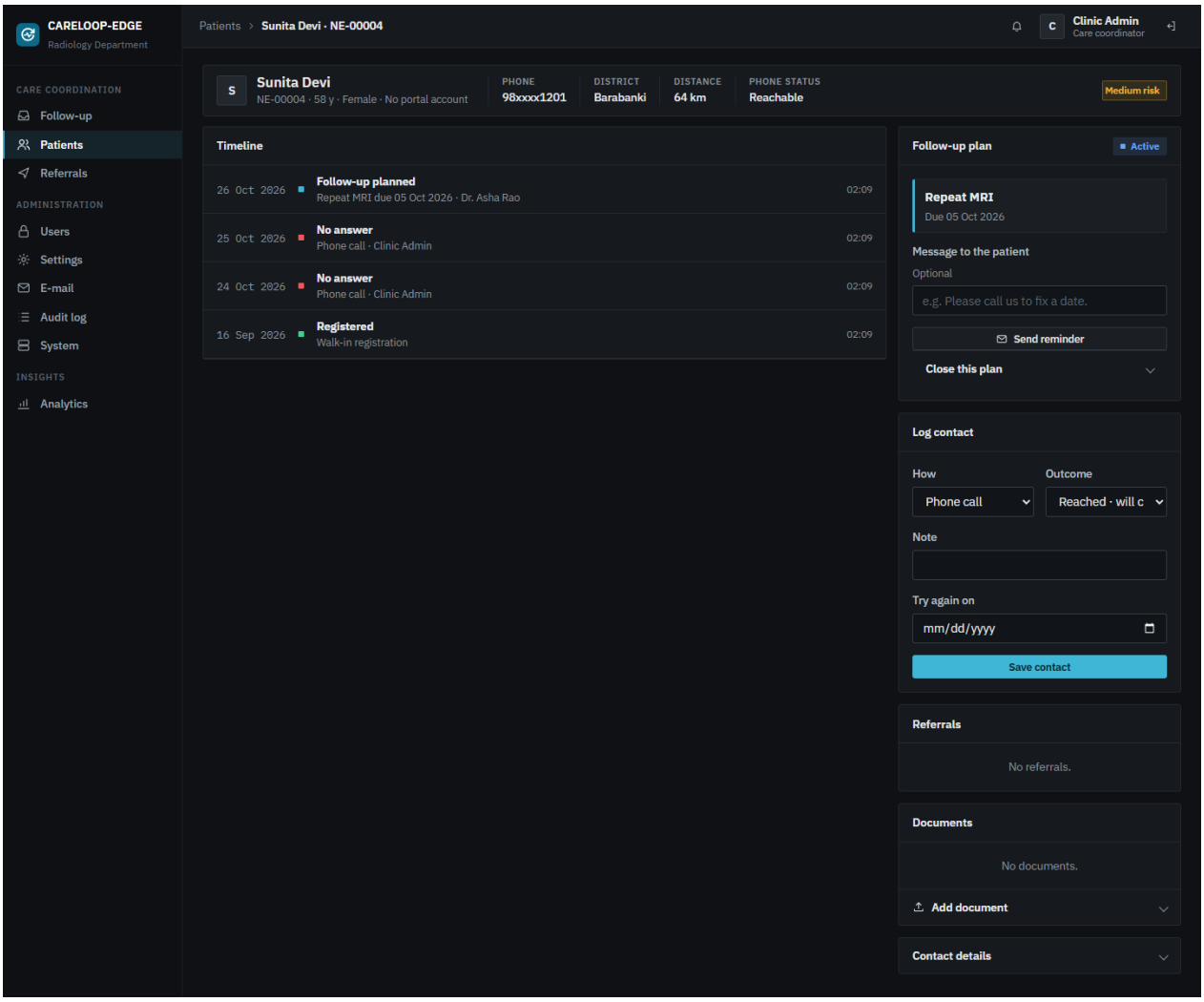


Figure 27: A patient record with the unified timeline, contact log, follow-up plan and referrals.

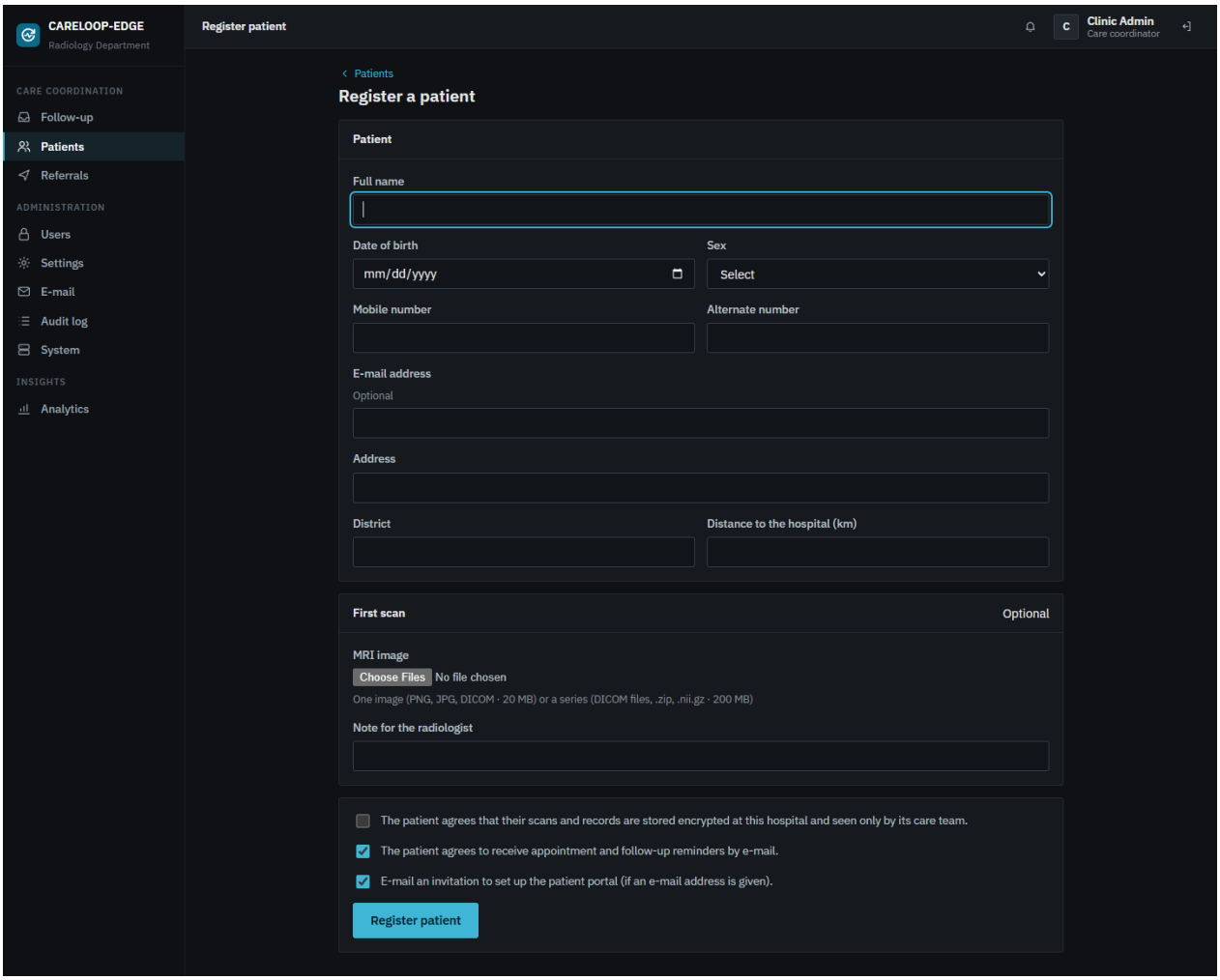


Figure 28: Walk-in registration at the desk, with a first scan; no e-mail address is needed.

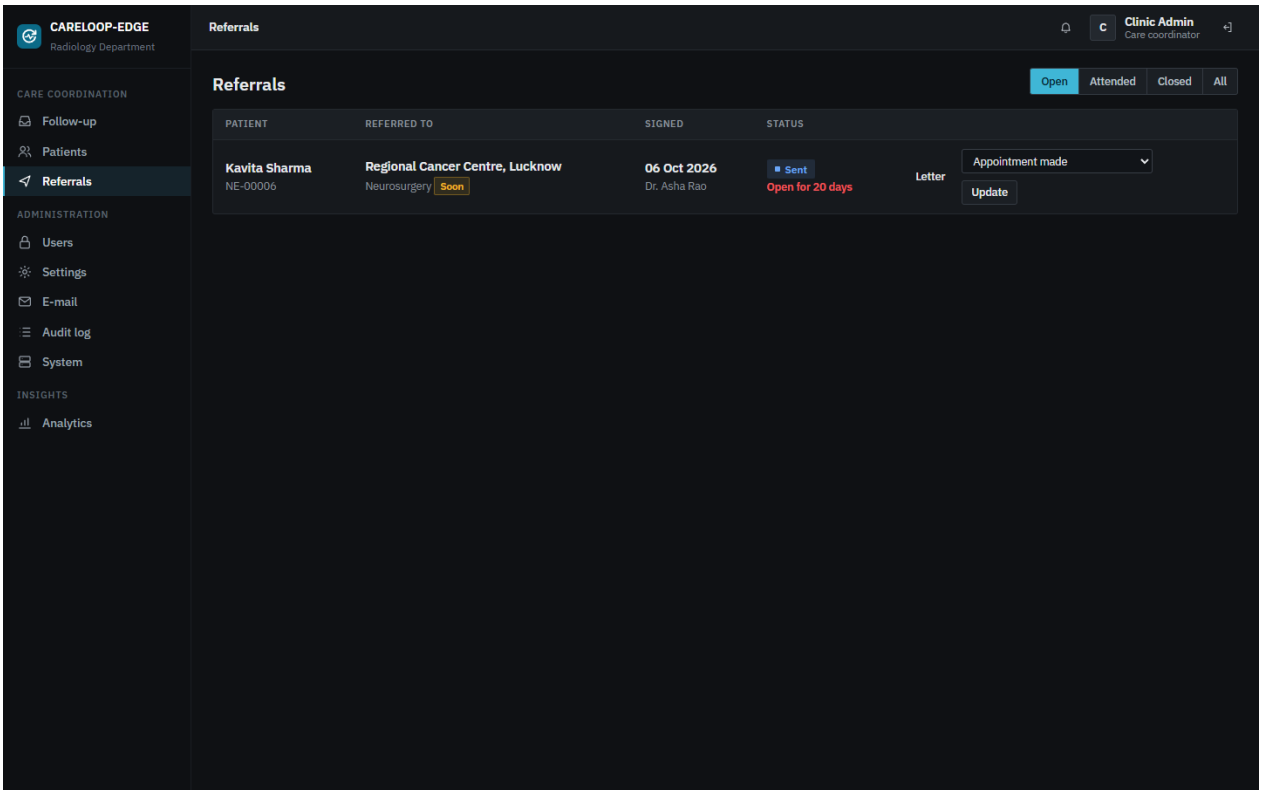


Figure 29: Referral tracking from signed letter to closed.

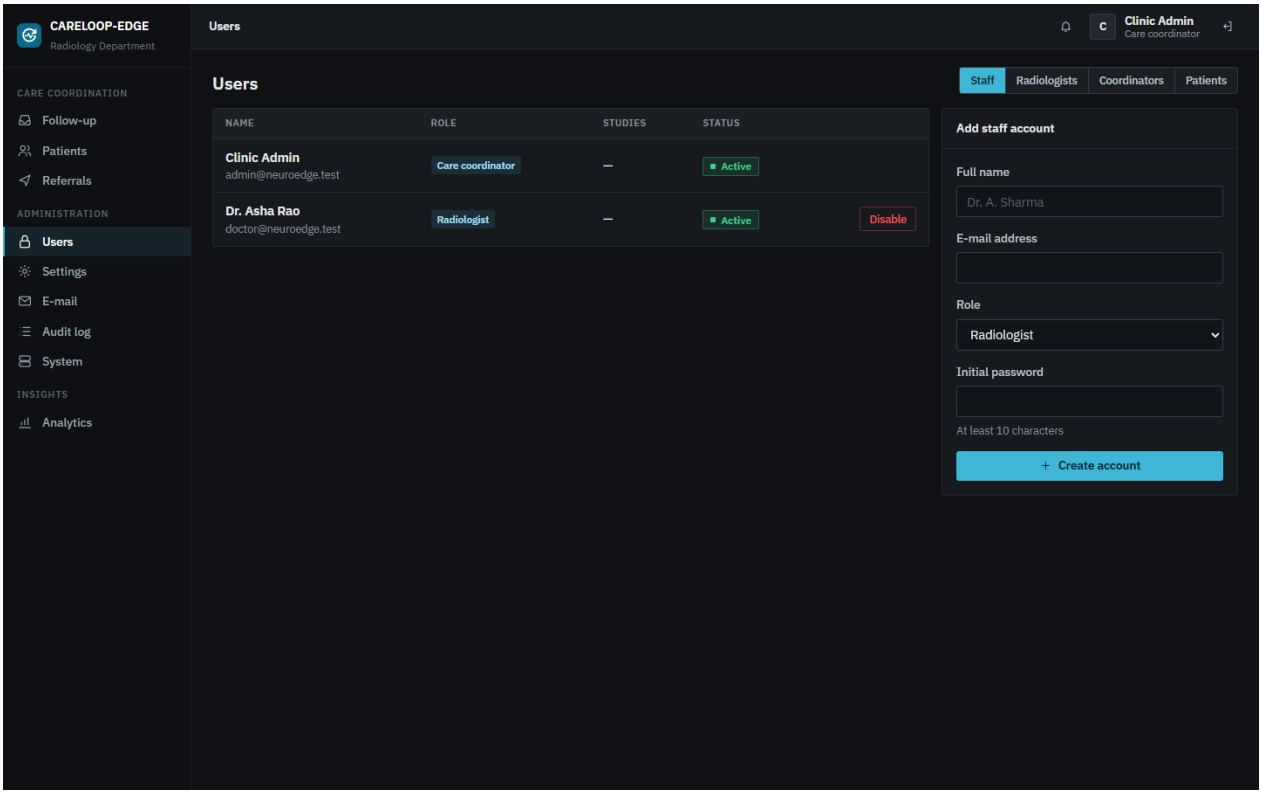


Figure 30: User management.

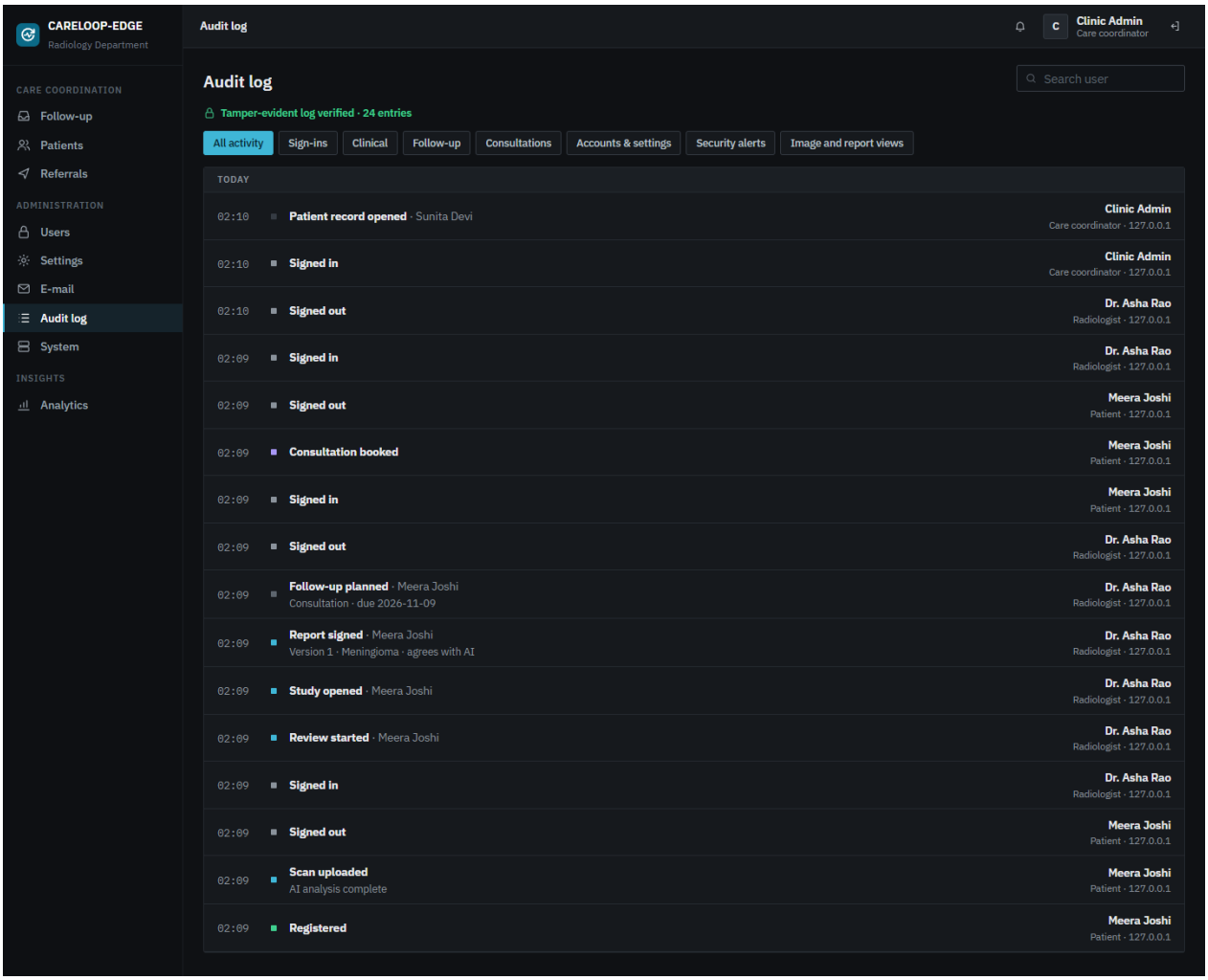


Figure 31: The tamper-evident audit log, verified on every view.

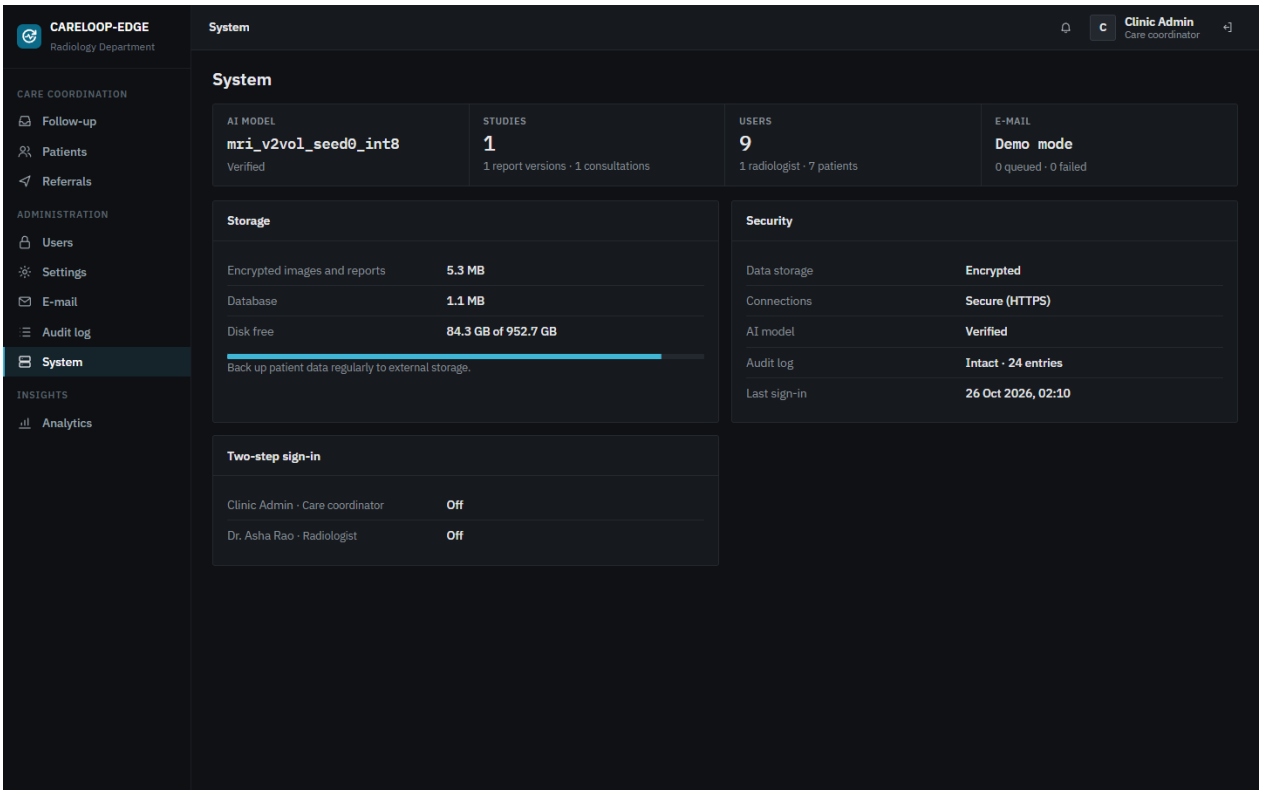


Figure 32: System status: model version and signature check, storage, security, audit chain.

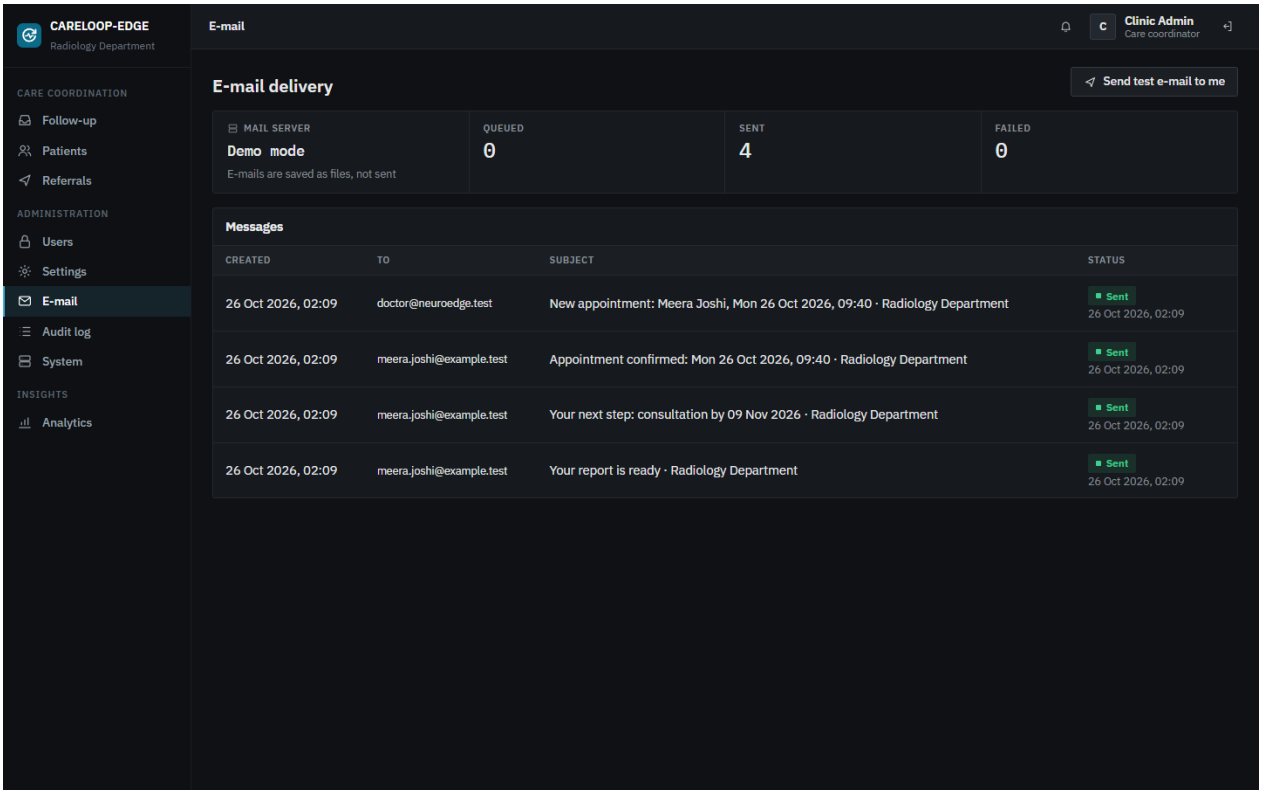


Figure 33: The e-mail outbox with delivery status and retries.

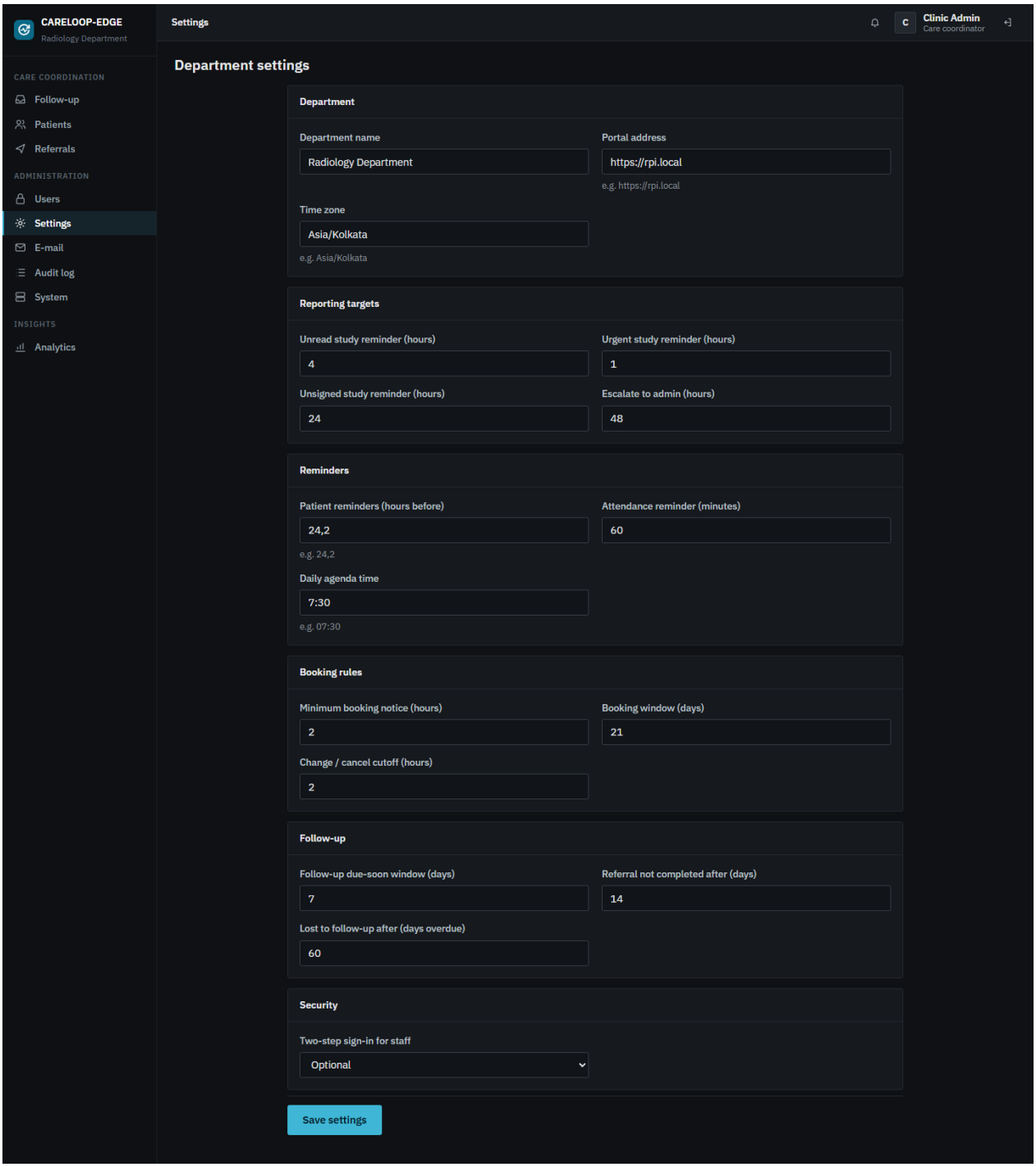
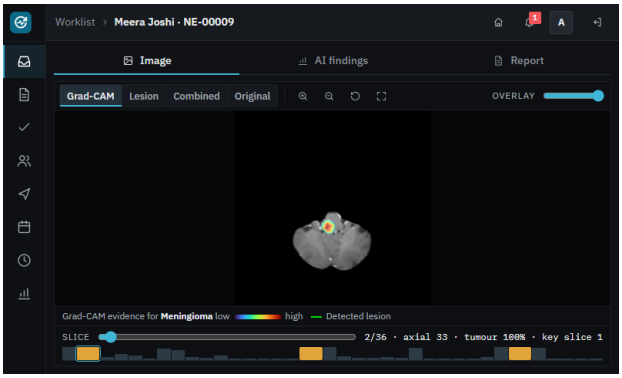
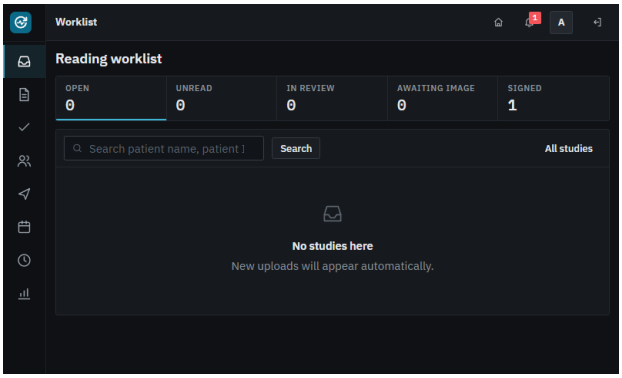


Figure 34: Department settings, including reminder and escalation timings.

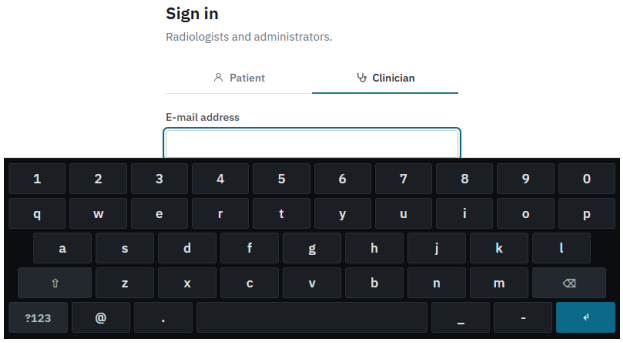
10.8 The device's own screen



(a) Case review as tabs.



(b) Worklist.



(c) Built-in touch keyboard.

Figure 35: Kiosk mode on the 7-inch display (800×480). The case page becomes Image / AI findings / Report tabs with the image filling the screen; touch targets are at least 44 px.

11 Device deployment

- **Raspberry Pi 4 (4 GB) with the official 7-inch display.** The clinic kiosk: portal, model `mri/0.3.0`, remote access through Tailscale Funnel. Installed from a newly written SD card (Raspberry Pi OS, first-boot configuration with key-only SSH), enrolled with the registry, model verified, kiosk running on the screen. The language model is not installed, which keeps memory free for whole-study analysis.
- **Kiosk.** A locked-down `kiosk` user without password, sudo or SSH logs in on the display and opens the device home in Chromium. Keyboard shortcuts that could leave the kiosk are disabled, and the screen returns to the device home after three idle minutes.

Problems found only on real hardware were fixed in the installer, for example SNI-less TLS by IP address, first-boot-only network configuration, the operating system's desktop auto-login overriding the kiosk, and a certificate tool waiting for a password prompt.

12 Testing and verification

- **163 automated tests** (pytest): CAM exactness, cryptography and tamper detection, leakage deduplication, portal roles and object-level access, CSRF and lock-out, the encrypted store, audit-chain verification, appointments, reminders, follow-up plans and risk score, whole-series upload and its safety checks (mixed patients or series refused), analytics figures on a seeded clinic, and the device page (no patient data).

- **Registry end-to-end check**: 16/16 against the live server.
- **Pi–PC parity**: 40/40 identical predictions.
- **Browser verification** at 800×480, 1280×720 and desktop sizes, with no Content Security Policy violations; the walk-through in Section 10 is itself an end-to-end test of the main workflow.

13 Limitations and future work

- Model v2 is one training seed; whole-study results are on held-out patients of the training datasets. The external BraTS-Africa test and two more seeds are pending.
- The healthy-volunteer set (IXI) is non-contrast T1; the 95% set is calibrated on slices, not on patients.
- Single-image prediction-set coverage is 0.91 against a 0.93 target; calibration per data source is planned.
- Measured Pi 4 whole-study latency is still to be recorded.
- BraTS data is licensed for research and non-commercial use.
- Planned: SMS and WhatsApp reminders in local languages, a PACS connection, and a pilot at a health centre.